Closing stomatal supply against photosynthetic demand at runtime: a verified diel plant growth model and the boundary of its diel starch balance

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ABSTRACT

Mechanistic models of plant growth usually prescribe carbon input or compute it from a photosynthesis routine that is solved separately from stomatal conductance. This work reports a modular simulation model in which the supply and demand sides of leaf gas exchange are closed against each other at every evaluation: a Jarvis conductance and a Farquhar assimilation function are combined into a quadratic in the intercellular CO_2 concentration and solved in closed form, replacing a fixed-point iteration that we show admits a period-two orbit under water stress. Around this closure the model integrates eleven states covering starch, sucrose and vacuolar sugar, nitrogen and phosphorus acquisition with explicit costs, and leaf and root biomass. Carbon conservation over the carbohydrate subsystem closes to machine precision. Rather than fitting coefficients per experimental condition, we hold a single parameter set across four photoperiods and report what that costs: relative growth rate is reproduced with a mean absolute percentage error of 8.58 % and a worst case of 11.75 %, against a legacy per-photoperiod baseline whose apparent agreement is a bisection residual of eight coefficients fitted to eight observations. Leave-one-photoperiod-out cross-validation gives a held-out error of 13.18 % for a photoperiod-independent growth coefficient and 4.36 % when the coefficient is allowed a linear photoperiod dependence. Diel starch requires only a near-unity multiplicative alignment, between 0.86 and 1.15, whereas sucrose requires a factor of two to three, which we report rather than absorb. Ablation shows that computing the carbon input, rather than prescribing it, is what carries the photoperiod response. Variance-based sensitivity analysis attributes only 20 % of output variance to parameters acting alone, placing the remainder in interactions rather than main effects. Finally we locate one boundary of the model's diel starch balance: under the reference chamber forcing the diel starch balance returns to its dawn value up to a 15 h photoperiod, whereas from 16 h the pool gains carbon on every cycle and still drifts upward after 90 simulated days. Under the higher light dose of the shipped configuration the balance does not return at any photoperiod tested, so the boundary is a property of the forcing as well as of the photoperiod. We characterise this limitation rather than conceal it.

1. Introduction

Arabidopsis thaliana grows in a light environment that switches off every day. During the light phase the plant assimilates carbon and stores part of it as transient leaf starch. During the night this reserve is the only carbon source for respiration, maintenance and growth. Measurements show that the reserve is degraded at an almost constant rate and is nearly exhausted at dawn [20, 51]. The plant therefore behaves as if it rations a known amount of carbon over a night of known length [45]. When the photoperiod changes, both the amount and the length change together, and the rationing problem changes with them [54, 53]. This makes photoperiod a useful test case for a simulation model, because a single environmental variable moves the light-phase carbon gain and the dark-phase reserve use at the same time.

The problem is difficult to simulate because it couples processes on very different time scales. Stomatal opening and leaf assimilation respond within minutes to light, temperature and humidity. Starch and sucrose pools turn over within hours. Nutrient status changes over days and biomass accumulates over weeks. The carbon available for growth at any instant is not a property of any single equation. It is the result of the environment, the gas-exchange machinery and the internal carbohydrate and nutrient state acting together [15, 57]. A model that describes only photosynthesis,

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or only biomass accumulation, cannot represent this. For a simulation study the main task is therefore to build a stable numerical workflow in which these subsystems exchange fluxes consistently over the day–night cycle.

Existing models resolve parts of this chain, but rarely the whole chain at the resolution the problem needs. Crop and field-scale simulators integrate growth, water and nitrogen over seasons, but their standard configurations prescribe the carbon input and advance the state on a daily step [64, 22, 26]. Leaf gas-exchange models resolve the process that a prescribed carbon input replaces, but are formulated and tested as steady-state systems at a fixed carbohydrate status, so they stop at the leaf [15, 25, 11]. Few published models compute the supply–demand operating point at every step and then carry that carbon input through a diel starch and sucrose economy into biomass. Section 2 develops this gap in detail.

This paper presents such a model and reports its evaluation. The model is a whole-plant dynamic simulator with eleven state variables, organised as six submodels with declared inputs, outputs and parameters (Figure 1). S1 maps clock time to the environmental drivers. S2 computes stomatal conductance from a multiplicative Jarvis formulation, which acts as the supply side of gas exchange. S3 computes net assimilation as a function of intercellular CO_2 from the Farquhar C3 biochemistry, which acts as the demand side. S4 closes supply against demand. S5 holds the diel carbohydrate economy, the starch and sucrose pools and the nitrogen and phosphorus uptake and costs. S6 converts sucrose into leaf and root biomass. Submodels S2–S4 form a fast algebraic subsystem, and S5–S6 form the slow differential subsystem that is integrated in time.

The methodological content sits in the runtime solution and the evaluation design, and this is what we expect to transfer to other coupled biophysical simulators. The gas-exchange operating point is solved algebraically at runtime: substituting Fick's law into the branch-wise Farquhar expression gives a quadratic in the intercellular CO_2 concentration, solved branch by branch with an explicit admissibility rule for the roots, so no inner iteration is required inside the derivative evaluation. The algebraic subsystem is then eliminated exactly inside each derivative call, and the remaining differential system is integrated with an explicit adaptive method. The implementation is verified before it is compared with data, and model behaviour is analysed with a variance-based global sensitivity analysis over declared parameter ranges. The contributions are listed at the end of this section.

The evaluation design keeps calibration and prediction separate. Four coefficients are fitted and shared across all photoperiods, in place of a legacy per-photoperiod calibration that left no degrees of freedom (Section 5). Predictive accuracy is then measured by holding out whole photoperiods: fitting on three light regimes and predicting the fourth. Under this design the model reaches a held-out mean absolute percentage error of 4.36 % in relative growth rate, when one growth coefficient is allowed to depend linearly on photoperiod (Figure 3). We also report the photoperiod range over which the model remains usable, together with the mechanism that breaks first.

The contributions of the paper are the following.

1. A coupled whole-plant simulation model in which Jarvis stomatal conductance and Farquhar C3 photosynthesis produce the carbon input at every step, and that input is propagated through diel starch and sucrose dynamics, nutrient uptake and costs, and biomass growth (Section 3).
2. An analytic runtime solution of the supply–demand closure as a quadratic in the intercellular CO_2 concentration, embedded in a fast/slow numerical workflow (Sections 3 and 4).
3. A verification protocol for the implementation that is independent of any comparison with data, covering the algebraic closure, the time integration and the carbohydrate bookkeeping (Section 4).
4. An evaluation with four shared fitted parameters instead of a per-photoperiod calibration. The design uses leave-one-photoperiod-out cross-validation, an ablation of the carbon-input pathway, and a global sensitivity analysis over declared ranges (Sections 5 and 6).
5. A measured validity envelope. We report the photoperiod range in which the diel starch balance closes, the photoperiod at which it stops closing, and the bound on the simulated growth response to day length (Sections 6 and 7.6).

The rest of the paper is organised as follows. Section 2 reviews mechanistic growth models, gas-exchange formulations, diel carbohydrate modelling and simulation evaluation practice, and states the gap that this work addresses. Section 3 gives the model formulation as six submodels with their states, inputs and parameters. Section 4 describes the numerical implementation and the verification results. Section 5 defines the photoperiod scenarios, the calibration and cross-validation design, the ablation variants and the sensitivity protocol. Section 6 reports the results, Section 7 interprets them, Section 7.6 states what cannot be concluded, and Section 8 concludes.

2. Related Work

This section positions the model against four bodies of work: mechanistic whole-plant and crop simulation, leaf gas-exchange modelling, diel carbohydrate turnover in *Arabidopsis thaliana*, and evaluation practice in simulation studies. The aim is to locate the contribution, not to survey each field.

2.1. Mechanistic plant growth simulation models

Mechanistic growth models link physiological processes, microclimate and biomass change over time, at scales from the leaf to the field and from minutes to whole seasons. Crop simulation platforms such as APSIM, DSSAT and WOFOST are mature tools for growth, water and nitrogen dynamics at field scale [22, 26, 64], and process-based formulations follow seasonal climate variation better than empirical ones [42, 32]. Other families address the same problem differently: functional–structural models combine architecture with physiology [30, 50, 12], soil–plant water transport models describe flow from soil through xylem [24], optimality models assume allocation maximises a criterion [43], and continuous formulations describe biomass or vegetation dynamics as differential equations [37, 57, 33, 17, 56].

Two properties of this literature matter here. Modular organisation is common and useful, because it lets each component be tested on its own [50, 68]. In the configurations most often used at field scale, the carbon input is prescribed rather than computed, through light-use-efficiency or radiation-use coefficients with a daily step [26, 64, 22]. That default separates the carbon source from the environment and from the internal state of the plant, so the model cannot represent the feedback between stomatal behaviour, intercellular CO_2, carbohydrate status and growth. This describes the usual configuration, not the capability of these platforms.

2.2. Photosynthesis and stomatal conductance modelling

For C3 plants the standard biochemical model is the Farquhar–von Caemmerer–Berry formulation, which relates net assimilation to intercellular CO_2 through Rubisco-limited carboxylation and RuBP regeneration [15, 6]; its parameters and their temperature dependence are well documented [48, 4, 13]. Assimilation also depends on the diffusive supply of CO_2 set by stomatal conductance [28]. The main conductance formulations are the empirical Jarvis model, the Ball–Berry family and the Medlyn optimality model [25, 2, 35]. The Jarvis form expresses conductance as a product of response functions to irradiance, vapour-pressure deficit, temperature, CO_2 and plant water status [25, 66, 67], and is convenient in simulation because time-varying drivers enter it directly [11, 5].

Coupling the two sides introduces a feedback: intercellular CO_2 sets the assimilation rate, while conductance sets how much CO_2 can diffuse in. The operating point is defined implicitly and must be found at every evaluation [11]. Iterative solutions are widely used, and analytic solutions for particular stomatal formulations are known [5, 27]. The Farquhar and Jarvis equations used here are inherited physiological components, not a contribution of this paper. The contribution is how they are solved at runtime and coupled to the carbohydrate, nutrient and biomass states.

2.3. Diel carbohydrate turnover and photoperiod

Carbohydrate dynamics link assimilation to growth. In the light, assimilated carbon is partitioned between mobile sucrose and transient leaf starch; in the dark, starch degradation supplies carbon for respiration, maintenance and growth [53, 51]. In *Arabidopsis thaliana* this degradation is closely regulated and the reserve is almost completely used by dawn [20, 45, 62]. Models of the process connect circadian regulation, sucrose availability and sugar signalling to partitioning and growth [18, 46], and the separation between cytosolic and vacuolar sucrose has been described in detail [29, 40].

Photoperiod changes the balance between daytime accumulation and nocturnal consumption, and *Arabidopsis thaliana* adjusts partitioning and growth to day length [54, 36, 1]. This makes photoperiod scenarios a suitable evaluation setting: they perturb both phases of the diel carbon economy through one variable, with documented responses. Growth also depends on nutrient status, and nitrogen and phosphorus enter mechanistic models as internal states or limiting factors [19, 65, 3, 49, 9]. Models that resolve the diel cycle in detail, however, usually treat assimilation as given [45, 20, 18], so the day–night balance and the gas exchange that produces it are not solved as one system.

2.4. Verification, validation and sensitivity practice in simulation

Two distinctions from simulation methodology are relevant here. The first is between verification and validation: agreement with measurements does not show that the numerical procedures inside the simulator are correct [68, 10], so residual checks against analytic solutions, integration refinement and bookkeeping checks address a different question

from data agreement. The second is between calibration residual and predictive error: when a model is fitted and evaluated against the same small set of aggregate observations, the reported error is a calibration residual, and reported accuracy depends strongly on the calibration protocol [31, 61, 22, 26]. Ablation attributes behaviour to model structure by removing, simplifying or replacing a component [68]. Variance-based sensitivity methods give first- and total-order indices and quantify interaction structure [52, 44]; Morris screening indicates nonlinearity and interaction more cheaply [39, 8], Latin hypercube designs are standard for sampling [34], and testing the estimator against a function with closed-form indices is good practice [23]. These indices describe model behaviour over the declared parameter domain; they do not establish physiological importance, and screening measures are not an exact variance decomposition.

2.5. Open problems addressed here

Read together, this literature leaves a specific gap. Leaf-level gas-exchange models solve the supply–demand operating point accurately, but at fixed carbohydrate status and without downstream dynamics [15, 25, 6]. Whole-plant and crop simulators carry growth over days to seasons, but prescribe or simplify the carbon input and rarely resolve the diel cycle [26, 64, 22]. Dedicated diel carbohydrate models resolve the day–night starch balance, but take assimilation as an input [45, 20]. Models that do compute assimilation biochemically and carry it into leaf-area growth are the closest published relatives [63], but they do not carry the diel starch and sucrose pools as integrated states. The feedback loop connecting conductance, intercellular CO_2, carbohydrate status and growth is therefore opened somewhere in each family. Two further problems are methodological. The algebraic constraint created by the coupling must be solved at every step, yet the solver and its behaviour outside the calibrated range are often left unstated. Calibration is also frequently performed against the same aggregate observations used for evaluation, so predictive error cannot be separated from a calibration residual.

This paper closes part of that gap. We do not propose new process equations for photosynthesis, stomatal conductance or starch turnover; those are taken from the literature. We add the coupled formulation with a computed carbon input, its analytic runtime closure and fast/slow numerical treatment, a verification protocol independent of data agreement, and an evaluation design in which whole photoperiods are held out and the fitted parameter set is shared across conditions. The aim is not to replace crop simulators or functional–structural models, but to provide a compact and reproducible core in which gas exchange is propagated explicitly into downstream states.

We also state where the model falls short, with numbers in Section 6. The model does not generate the full observed photoperiod dependence of growth from its shared dynamics; part of that dependence remains localised in one growth-conversion coefficient. The diel starch balance closes only up to an upper photoperiod, above which the reserve accumulates from day to day (Figure 7). Canopy geometry, water transport, temperature acclimation and multi-species parameterisation are outside the present scope. These are stated as named structural limitations in Section 7.6.

3. Model Formulation

This section states the model as it is implemented. We fix the system boundary and the decomposition into submodels, give the state vector, and then specify each submodel in the order in which information flows through one derivative evaluation. Every equation corresponds to a function in the released C source, and every parameter value is the value used in the reported runs. Notation is introduced once here and used unchanged in the rest of the paper. Table 1 collects the states, inputs and derived quantities, Table 2 the parameters with their provenance, and Table 3 the four calibrated coefficients.

3.1. Conceptual model and system decomposition

The system is a single vegetative *Arabidopsis* plant with its immediate physical environment. Radiation, air temperature, humidity, ambient CO_2, leaf water potential and soil nutrient concentrations cross the boundary as external inputs. Carbon and nutrient acquisition, storage and allocation to leaf and root biomass are internal states. The plant is two mass compartments, leaf and root, without explicit geometry, so internal transport is described by aggregate pools. This lumped decomposition and the carbon–nutrient allocation states follow the whole-plant growth framework of Tedone [55], whose prescribed carbon input is here replaced by a computed gas-exchange layer.

Figure 1 shows the decomposition into six submodels and the signals they exchange.

S1 *Environmental forcing* maps clock time to the light switch and the micro-climate. *In:* t, scenario parameters. *Out:* PAR, T, RH, VPD, C_a, ψ_w, L.

S2 *Stomatal conductance (supply)*, a multiplicative Jarvis model. *In:* PAR, VPD, T, C_a, ψ_w. *Out:* g_s.

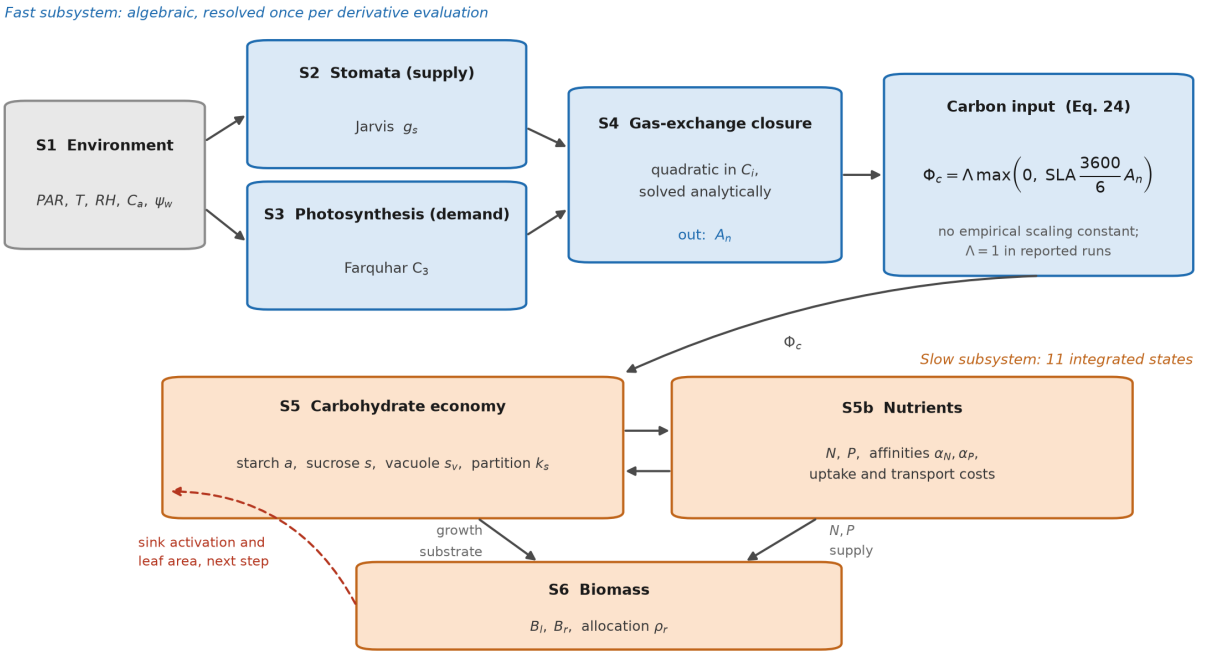


Figure 1: Model architecture and information flow through one derivative evaluation. The fast subsystem (blue) is algebraic and is resolved once per evaluation: environmental forcing S1 drives the stomatal supply function S2 and the photosynthetic demand function S3, which are closed against each other in S4 as a quadratic in the intercellular CO_2 concentration and solved in closed form. The resulting net assimilation is converted to the model carbon input Φ_c by Eq. (22), which the figure reproduces so that the conversion can be read without turning back; there is no empirical scaling constant in it. The slow subsystem (orange) comprises the eleven integrated states: the carbohydrate economy S5, nutrient acquisition with its explicit costs S5b, and leaf and root biomass S6. The dashed return arrow is the only slow-to-fast signal in the architecture and it is narrow: leaf biomass enters the gas-exchange layer solely as a viability gate, which returns zero assimilation below $B_l = 0.001$ gFW and has no effect above it. Sink activation is a feedback within the carbohydrate economy, driven by the sucrose pool and not by biomass, so it is drawn inside S5. Arrows show the direction of information flow, not stoichiometric mass transfer; the carbon budget is stated in Section 4.3.

S3 *Leaf photosynthesis (demand)*, the Farquhar C_3 model. *In:* PAR, C_i, $V_{c\,\max}$, $J_{\max}$. *Out:* $A_n(C_i)$.

S4 *Gas-exchange coupling* closes supply against demand. *In:* g_s, $A_n(C_i)$, C_a. *Out:* A_n, C_i.

S5 *Carbohydrate and nutrient economy* holds the diel starch and sucrose pools, the vacuolar compartment, and sucrose-limited N and P uptake with its costs. *In:* Φ_c, soil N_s, P_s. *Out:* s, U_N, U_P.

S6 *Growth and allocation* converts exported sucrose into leaf and root biomass under functional-balance control. *In:* s, α_N, α_P, ς. *Out:* B_l, B_r, ρ_r.

Submodels S2–S4 form a fast algebraic subsystem. Leaf gas exchange equilibrates within seconds, so it is treated as instantaneously relaxed relative to the slow states in S5 and S6, which evolve over the diel cycle. It is therefore not integrated but evaluated in closed form at every derivative call, as an algebraic constitutive layer inside the derivative of the slow states [10, 68]. This separation organises both the formulation and the implementation.

Only one signal returns from the slow subsystem to the fast one: leaf biomass B_l acts as a viability gate, and the carbon input is zero when $B_l < B_{l,\min}$. The photosynthetic capacities $V_{c\,\max}$ and $J_{\max}$ are fixed parameters and do not respond to nitrogen or carbohydrate status. Figure 1 must therefore not be read as showing a capacity feedback that the equations do not contain.

The formulation rests on six assumptions.

(A1) The plant is lumped into leaf and root compartments; no within-organ gradients or canopy architecture are resolved.

(A2) Leaf gas exchange is at steady state at every instant, so intercellular CO_2 is an algebraic balance and not a state.
(A3) Stomatal conductance responds multiplicatively to the environment and not to the assimilation rate [25, 11].
(A4) Transitory starch is mobilised at a nocturnal rate paced by night length, as the observed near-linear depletion indicates [20, 45].
(A5) Active nutrient uptake is limited by cytosolic sucrose, so carbon and nutrient acquisition are co-limiting [51].
(A6) Partitioning follows functional-balance theory: investment shifts towards the most limiting resource [56, 57].

3.2. State-space representation

The slow states form the vector $X(t) \in \mathbb{R}^{11}$ governed by

$$\frac{dX(t)}{dt} = f(X(t),\, U(t),\, \boldsymbol{\theta},\, t)\,, \tag{1}$$

where $U(t)$ is the exogenous forcing and $\boldsymbol{\theta}$ the time-invariant parameter set. The explicit time dependence enters only through the diel forcing. In the order used by the state packing routine of the implementation,

$$X(t) = \left[k_s,\ a,\ s,\ \alpha_N,\ \alpha_P,\ N,\ P,\ \rho_r,\ B_l,\ B_r,\ s_v\right]^{\mathsf{T}}. \tag{2}$$

Each symbol is defined in Table 1. Total biomass $B = B_l + B_r$ is recomputed from the two biomass states whenever it is needed and is not integrated as a twelfth state.

The pool states a, s, s_v, N and P are normalised model pools on a nominal per-fresh-weight basis, not concentrations of a growing body. We adopt this reading because it is the one under which the implemented balance equations are exact: no growth-dilution term appears in Eqs. (25), (27), (28), (36) or (37). The cost is that a and s are not directly measured tissue concentrations, so the carbohydrate comparison in Section 6 is a comparison of shape and timing after multiplicative alignment, not of absolute level.

The forcing vector is

$$U(t) = \left[PAR(t),\ T(t),\ RH(t),\ VPD(t),\ C_a,\ \psi_w,\ L(t)\right]^{\mathsf{T}}, \tag{3}$$

The deficit VPD is a diagnostic function of T and RH through Eq. (6) and is not an independent input; it is listed because it enters the stomatal response directly. In the scenarios reported here RH, C_a and ψ_w are constant; T is either constant or thermoperiodic, and the chamber scenarios against which the model is evaluated hold 21 °C in the light and 19 °C in darkness. The time-varying inputs are therefore PAR, L and, under the chamber protocol, T.

3.3. Submodel S1: environmental forcing

Let $h(t) = t \bmod 24$ be the hour of day and f the photoperiod length in hours. The light switch is

$$L(t) = \begin{cases} 1, & 0 \le h(t) < f, \\ 0, & f \le h(t) < 24, \end{cases} \tag{4}$$

and the incident radiation follows one of two daytime profiles, selected by scenario:

$$PAR(t) = \begin{cases} PAR_{\max}, & \text{square wave, } L = 1, \\ PAR_{\max}\,\sin(\pi\, h(t)/f)\,, & \text{half sine, } L = 1, \\ 0, & L = 0. \end{cases} \tag{5}$$

The square wave represents a growth chamber, which delivers constant irradiance over the light period; the half sine is a smooth idealisation of a natural light course. The choice sets the daily light integral: the square wave gives $PAR_{\max} f$ and the half sine $(2/\pi)\, PAR_{\max} f$. A run matching a chamber experiment must therefore use the square

Table 1
State variables, environmental inputs and the principal derived quantities. Pool states are normalised model pools on a nominal per-fresh-weight basis (Section 3.2). "Derived" quantities are recomputed algebraically at every derivative evaluation and are not integrated.

Symbol	Meaning	Unit	Role
k_s	Starch partition coefficient	–	state, Eq. (24)
a	Transitory starch pool	μmol C_6 gFW^{-1}	state, Eq. (25)
s	Cytosolic sucrose pool	μmol C_6 gFW^{-1}	state, Eq. (27)
s_v	Vacuolar sucrose pool	μmol C_6 gFW^{-1}	state, Eq. (28)
α_N	Nitrogen investment state	–	state, Eq. (39)
α_P	Phosphorus investment state	–	state, Eq. (40)
N	Internal nitrogen pool	μmol N gFW^{-1}	state, Eq. (36)
P	Internal phosphorus pool	μmol P gFW^{-1}	state, Eq. (37)
ρ_r	Sucrose-to-root allocation state	–	state, Eq. (44)
B_l	Leaf biomass	gFW	state, Eq. (42)
B_r	Root biomass	gFW	state, Eq. (43)
PAR	Photosynthetically active radiation	μmol m^{-2} s^{-1}	input, Eq. (5)
L	Light/dark switch	–	input, Eq. (4)
T	Leaf temperature	°C	input, constant
RH	Relative humidity	%	input, constant
C_a	Ambient CO_2 concentration	μmol mol^{-1}	input, constant
ψ_w	Leaf water potential	MPa	input, constant
N_s, P_s	Soil N, P concentration	μmol cm^{-3}	input, constant
VPD	Vapour pressure deficit	kPa	derived, Eq. (6)
g_s, g_c	Conductance to H_2O, to CO_2	mol m^{-2} s^{-1}	derived, Eqs. (7), (17)
C_i	Intercellular CO_2 concentration	μmol mol^{-1}	derived, Eq. (21)
A_c, A_j	Rubisco-, RuBP-limited rate	μmol m^{-2} s^{-1}	derived, Eqs. (13), (14)
A_n	Net leaf assimilation	μmol m^{-2} s^{-1}	derived, Eq. (15)
J	Electron transport rate	μmol m^{-2} s^{-1}	derived, Eq. (16)
Φ_c	Carbon input to the pools	μmol C_6 gFW^{-1} h^{-1}	derived, Eq. (22)
D_a	Starch degradation flux	μmol C_6 gFW^{-1} h^{-1}	derived, Eq. (26)
E	Exportable sucrose term	μmol C_6 gFW^{-1}	derived, Eq. (29)
U_N, U_P	Realised N, P uptake	μmol gFW^{-1} h^{-1}	derived, Eq. (35)
ς	Stoichiometric signal	–	derived, Eq. (45)
B	Total biomass, $B_l + B_r$	gFW	derived

profile at the chamber irradiance. One consequence of the design affects every photoperiod result. With $PAR_{\max}$ held fixed, changing the photoperiod changes the daily light integral in proportion. Day length and daily photon dose are confounded by construction, so a photoperiod response in this model is a joint response to duration and dose.

The vapour pressure deficit is obtained from leaf temperature and relative humidity through the Tetens relation [38, 41],

$$VPD = e_s\left(1 - \frac{RH}{100}\right), \qquad e_s = 0.61078\,\exp\!\left(\frac{17.27\,T}{T + 237.3}\right), \tag{6}$$

with e_s and VPD in kPa and T in °C. Where a thermoperiod is specified, T takes its day value while $L = 1$ and its night value while $L = 0$; otherwise T is constant.

3.4. Submodel S2: stomatal conductance (supply)

This submodel returns the stomatal conductance and forms the supply side of gas exchange. It carries no dependence on the assimilation rate (A3). Conductance is a maximal value reduced by a product of dimensionless environmental modifiers, each bounded in [0, 1] [25, 11]:

$$g_s = g_{s,\max}\, f_{PAR}(PAR)\, f_{VPD}(VPD)\, f_T(T)\, f_{CO_2}(C_a)\, f_w(\psi_w). \tag{7}$$

The multiplicative closure assumes that the environmental controls act independently. It is a widely used empirical conductance model and needs no optimisation step at runtime [11, 5]. The maximal conductance $g_{s,\max}$ is a separate parameter from the shape parameters of the modifiers. Each modifier can then be required to equal unity at its own non-limiting reference condition, so changing the shape of one modifier does not rescale the whole conductance.

The light response is a saturating hyperbola with a residual conductance b_{10} below a breakpoint $q = b_{10}/b_1$,

$$f_{PAR} = \begin{cases} b_{10}, & PAR \le q, \\ \dfrac{b_1 b_2\,(PAR - q)}{b_1 + b_2\,(PAR - q)}, & PAR > q. \end{cases} \tag{8}$$

The response to atmospheric demand is linear and floored at zero, and the response to leaf water potential is a saturating recovery above a closure potential $\psi_{\min}$, clipped to $[0, 1]$:

$$f_{VPD} = \max\bigl(0,\ 1 - b_5\, VPD\bigr), \tag{9}$$

$$f_w = 1 - \exp\bigl(-b_6\,(\psi_w - \psi_{\min})\bigr). \tag{10}$$

The temperature response is a skewed optimum with zeros at T_1 and T_h,

$$f_T = \begin{cases} b_3\,(T - T_1)\,(T_h - T)^{b_4}, & T_1 < T < T_h, \\ 0, & \text{otherwise}, \end{cases} \qquad b_4 = \frac{T_h - T_0}{T_0 - T_1}, \quad b_3 = \frac{1}{(T_0 - T_1)(T_h - T_0)^{b_4}}. \tag{11}$$

The two constants b_3 and b_4 are not free. They are fixed by requiring stationarity at T_0 and $f_T(T_0) = 1$, which places the maximum at T_0 and normalises the modifier to unity there. With $T_1 = 5$, $T_0 = 25$ and $T_h = 45\,^\circ\mathrm{C}$ this gives $b_4 = 1$, so the curve is a single-peaked parabola and no clipping from above is needed.

The CO_2 response is piecewise linear in C_a, expressed in $\mu\mathrm{mol\,mol}^{-1}$. It is constant below 100, falls linearly to a floor b_8 at 1000, and equals that floor above it:

$$f_{CO_2} = \begin{cases} 1, & C_a < 100, \\ 1 - b_7\,(C_a - 100), & 100 \le C_a < 1000, \\ b_8, & C_a \ge 1000, \end{cases} \qquad b_7 = \frac{1 - b_8}{900}. \tag{12}$$

3.5. Submodel S3: leaf photosynthesis (demand)

This submodel returns net assimilation as a function of the intercellular CO_2 concentration and forms the demand side of gas exchange. It is the Farquhar–von Caemmerer–Berry C_3 model, in which gross assimilation is the smaller of a Rubisco-limited and an electron-transport-limited rate [15, 6]. The Rubisco-limited rate is

$$A_c = V_{c\max}\,\frac{C_i - \Gamma^*}{C_i + K_c\bigl(1 + O/K_o\bigr)}, \tag{13}$$

with Γ^* the CO_2 compensation point in the absence of dark respiration, K_c and K_o the Michaelis constants of Rubisco for CO_2 and O_2, and O the oxygen concentration. The electron-transport-limited rate is

$$A_j = J\,\frac{C_i - \Gamma^*}{4.5\,C_i + 10.5\,\Gamma^*}, \tag{14}$$

where the denominator coefficients 4.5 and 10.5 encode the electron requirement per carboxylation and per oxygenation under the standard stoichiometry of RuBP regeneration [7, 6]. Net assimilation is the limiting rate less dark respiration,

$$A_n = \min(A_c,\ A_j) - R_d. \tag{15}$$

The electron transport rate follows from PAR through the non-rectangular hyperbola $\theta J^2 - (J_{\max} + \phi\, PAR)J + J_{\max}\phi\, PAR = 0$ [15, 16], of which the lower root is taken,

$$J = \frac{\left(J_{\max} + \phi\, PAR\right) - \sqrt{\left(J_{\max} + \phi\, PAR\right)^2 - 4\,\theta\, J_{\max}\,\phi\, PAR}}{2\,\theta}, \tag{16}$$

with ϕ the apparent quantum yield for electron transport and θ the curvature of the light response.

No irradiance threshold is applied to this submodel: the carbon input is evaluated whenever $PAR > 0$ and $B_l \geq B_{l,\min}$. A threshold is unnecessary because, with the closed-form operating point of Section 3.6, the net rate passes continuously through zero at the light compensation point and the clamp in Eq. (22) returns exactly zero below it. Below that point A_n itself remains negative. It is the carbon input that is clamped and not A_n, so respiratory efflux is never counted as an assimilatory gain.

3.6. Submodel S4: gas-exchange coupling

Equations (13)–(15) give demand as a function of a concentration that is itself set by supply. Fickian diffusion of CO_2 through the stomata gives

$$A_n = g_c\,(C_a - C_i), \qquad g_c = \frac{g_s}{1.6}, \tag{17}$$

where the factor 1.6 converts the conductance to water vapour into a conductance to CO_2, which diffuses more slowly in air [6, 41]. Because A_n depends on C_i and C_i depends on A_n, the operating point is the mutual solution of Eqs. (15) and (17). How that coupling is solved is the main methodological choice in the formulation.

The system needs no iteration. Both Farquhar branches have the same rational form in C_i,

$$A = \frac{\mathrm{num}\,(C_i - \Gamma^*)}{d_1 C_i + d_0} - R_d, \qquad (\mathrm{num},\ d_1,\ d_0) = \begin{cases} \left(V_{c\max},\ 1,\ K_c(1 + O/K_o)\right), & \text{Rubisco branch,} \\ \left(J,\ 4.5,\ 10.5\,\Gamma^*\right), & \text{transport branch,} \end{cases} \tag{18}$$

so the branch coefficients are read directly off Eqs. (13) and (14) without dividing through. Equating supply and demand,

$$g_c\,(C_a - C_i)\,(d_1 C_i + d_0) = \mathrm{num}\,(C_i - \Gamma^*) - R_d\,(d_1 C_i + d_0), \tag{19}$$

and clearing the denominator gives a quadratic in C_i,

$$a\,C_i^2 + b\,C_i + c = 0, \qquad \begin{aligned} a &= -g_c\,d_1, \\ b &= g_c\,(d_1 C_a - d_0) - \mathrm{num} + R_d\,d_1, \\ c &= g_c\,d_0\,C_a + \mathrm{num}\,\Gamma^* + R_d\,d_0. \end{aligned} \tag{20}$$

Exactly one root of Eq. (20) is admissible on each branch, and no search is needed to identify it. The admissibility condition is $C_i \geq 0$ alone. For $g_c > 0$ the leading coefficient is $a = -g_c d_1 < 0$, while $c > 0$ because every factor in it is positive. The product of the roots c/a is therefore negative, so each branch has one positive and one negative root, and the positive one is the operating point:

$$C_i = \frac{-b - \sqrt{b^2 - 4ac}}{2a} \;>\; 0. \tag{21}$$

Substituting this root back into Eq. (18) gives the branch assimilation, and the smaller of the two branch values is returned. That is the same co-limitation rule as $\min(A_c, A_j)$ in Eq. (15), applied after each branch has been closed against supply rather than before. Because exactly one root per branch is admissible, the minimum over the two branches is the solution of the coupled system and not merely a candidate for it.

The admissibility rule deserves emphasis, because the natural guess is wrong. No upper bound on C_i is imposed, and a root above the ambient concentration is retained rather than rejected. By Eq. (17), $C_i > C_a$ is exactly what the supply relation returns whenever $A_n < 0$, that is, whenever the leaf is a net source and CO_2 diffuses outward. The condition $0 \le C_i \le C_a$ is therefore not a physical constraint here, and imposing it would discard the correct root twice a day.

The vanishing-conductance limit and the exactly-closed-stoma boundary are two distinct states, both with physical answers rather than clamps, and both are derived in Appendix A. In the production configuration the Jarvis conductance of Section 3.4 keeps a positive floor on its light response, so $g_c = 0$ is never reached and neither boundary case is exercised in any reported run.

Two properties of this closure matter for the rest of the paper. The branch-wise quadratic is solved once per derivative evaluation, with no inner iteration in the production configuration, so the derivative is a deterministic function of (X, t) as an embedded-pair error estimate assumes. And the solution is exact up to floating-point arithmetic, so no inner convergence tolerance interacts with the outer error control. Section 4 reports the substitution residual that verifies this.

3.7. From assimilation to model carbon input

The leaf-level net assimilation A_n (μmol CO_2 m^{-2} s^{-1}) must be converted to the carbon input Φ_c that drives the carbohydrate pools (μmol C_6 gFW^{-1} h^{-1}). The conversion is

$$\Phi_c = \Lambda \;\max\left(0,\; \underbrace{\mathrm{SLA}}_{\mathrm{m^2\,gFW^{-1}}} \;\; \underbrace{\frac{3600}{6}}_{\mathrm{s\,h^{-1}\ /\ mol\,CO_2\ per\ mol\,C_6}} \;\; A_n\right), \qquad L = 1, \tag{22}$$

and $\Phi_c = 0$ when $L = 0$ or $B_l < B_{l,\min}$. Three factors carry the whole unit conversion and none of them is free: SLA moves the flux to a per-fresh-weight basis, the factor 3600 converts seconds to hours, and the divisor 6 converts moles of CO_2 to moles of hexose. Their product is 600 SLA, so at the measured SLA of Table 2 the coefficient is fixed by the units and no empirical scaling constant enters the carbon-input pathway. The clamp keeps a negative A_n out of the pools, as discussed in Section 3.5.

The factor Λ is a sink-limitation term that down-regulates assimilation as the starch pool fills,

$$\Lambda = \min\left(1,\; \max\left(\lambda_c,\; \lambda_c + (1 - \lambda_c)\,\frac{a_{\max} - a}{a_{\max}}\right)\right), \qquad a_{\max} = D_{a,\max}\,(24 - f), \tag{23}$$

where λ_c is the residual assimilation fraction at a full pool and the ceiling $a_{\max}$ scales with night length, since a longer night requires a larger reserve. The inner max bounds the term at its value at the ceiling, so a pool above $a_{\max}$ cannot drive Λ negative and reverse the sign of assimilation. In all reported runs $\lambda_c = 1$, which makes $\Lambda = 1$ identically and the sink limitation inactive. The equation is given because the term is present in the right-hand side and is used in the ablation of Section 5; its shipped value is stated so that a reader knows the reported results do not depend on it.

3.8. Submodel S5: carbohydrate economy

The carbon input Φ_c is split between transitory starch and cytosolic sucrose by the partition coefficient k_s. This coefficient is a state with its own dynamics rather than a fixed split, so the plant can shift it during the day and again at night in response to the sucrose level. The diel carbohydrate balance is then an emergent property of the model and not a prescribed schedule [53, 36].

Starch partition. Let $\sigma^- = s_{\min}/(s_{\min}+s)$ and $\sigma^+ = s/(s+s_{\max})$ be the sucrose-deficit and sucrose-excess fractions, both in [0, 1] and both saturating. The partition state follows

$$\frac{dk_s}{dt} = \underbrace{L\left[-k_s\,\lambda_{sdr}\,\sigma^- + \left(1-k_s\right)\lambda_{sdi}\,\sigma^+\right]}_{\text{light phase}} + \underbrace{(1-L)\left(1-k_s\right)\lambda_{sni}\,\sigma^-}_{\text{dark phase}}. \tag{24}$$

Each term has a direct interpretation. In the light, a low sucrose level makes σ^- large and the first term reduces k_s, so more incoming carbon goes to sucrose and the immediate demand is met. A high sucrose level makes σ^+ large and the second term raises k_s, so surplus carbon is stored as starch. In the dark there is no input to partition, and the third term raises k_s when sucrose is low, returning the state to a storage-biased value before the next light period. The factors k_s and $(1-k_s)$ keep the state in [0, 1] without a clamp, since each term vanishes at the bound it approaches. The three rate coefficients set the speed of the day-reduction, day-increase and night-increase responses.

Starch pool. Starch gains the partitioned share of the input and loses the nocturnal degradation flux D_a,

$$\frac{da}{dt} = k_s\,\Phi_c - D_a. \tag{25}$$

The degradation flux implements the observed pacing of starch mobilisation against the remaining night length [20, 45]. Let a_{dusk} be the starch content recorded at the last transition to darkness and $a_{\min}$ the residual pool that is not mobilised. Then

$$D_a = (1-L)\min\left(D_{a,\max},\ \left[\frac{h}{24} + \left(1-\frac{h}{24}\right)\left(1-\frac{s}{s+s_{\max}}\right)\right]\frac{a_{\text{dusk}}-a_{\min}}{24-f}\right), \tag{26}$$

and $D_a = 0$ if $a_{\text{dusk}} < a_{\min}$. The quotient $(a_{\text{dusk}} - a_{\min})/(24-f)$ is the constant rate that would exactly exhaust the mobilisable reserve over the night, which is the rate the measurements show. The bracketed factor modulates it: when sucrose is abundant the second summand is small and degradation is slowed, so the reserve is spared. The outer min bounds the flux by the degradation capacity $D_{a,\max}$ and the factor $(1-L)$ confines it to darkness. Two properties matter for the limitation reported in Section 7.6. The reference level a_{dusk} is sampled at dusk and held over the night, so it is not a continuous function of the state, and the divisor $24-f$ is the nominal rather than the remaining night length.

Sucrose pools. Cytosolic sucrose receives the unpartitioned share of the input and the starch degradation flux, and pays four costs: nutrient uptake, phloem transport, maintenance respiration, and export to the sinks. It also exchanges with the vacuolar pool. The balance is

$$\frac{ds}{dt} = \left(1-k_s\right)\Phi_c + D_a - C_u - C_t - R_m - \eta\,E - k_{\text{in}}\,s + k_{\text{out}}\,s_v - C_g, \tag{27}$$

$$\frac{ds_v}{dt} = k_{\text{in}}\,s - k_{\text{out}}\,s_v. \tag{28}$$

The vacuolar compartment is a passive first-order buffer with influx rate k_{in} and efflux rate k_{out}, and the two rate constants appear with opposite signs in Eqs. (27) and (28), so the exchange conserves sucrose. It represents the measured separation of leaf sucrose into an active cytosolic fraction and a larger stored vacuolar fraction [29, 40]. Only s enters the cost and export terms, so the buffer damps the diel excursion of the active pool without adding or removing carbon, with relaxation time $1/(k_{\text{in}}+k_{\text{out}})$.

The export term is the phloem-loading flux, and it carries the sugar-induced sink activation:

$$E = \underbrace{\min\left(s,\ s_{\max}\right)}_{\text{loadable pool}} \underbrace{\left[1 + \left(\frac{s}{K_s}\right)^{n_H}\right]}_{\text{sink activation}} \underbrace{\left[\lambda_g + \left(1-\lambda_g\right)\left(1 - L + L\left(1-k_s\right)\right)\right]}_{\text{loading efficiency}}. \tag{29}$$

The activation factor scales the export flux, not the pool size, and is deliberately unbounded rather than a Hill function. A bounded factor $s^{n_H}/(K_s^{n_H} + s^{n_H})$ can only weaken the drain, which is the wrong direction: sucrose induces its own consumption by raising invertase activity and sink-tissue transporter expression, so consumption capacity rises with the pool [51, 53]. The unbounded factor $1 + (s/K_s)^{n_H}$ represents this, and holds the steady-state pool near K_s while assimilation varies, which a first-order drain with constant coefficients cannot do. The half-activation constant K_s is one of the four calibrated parameters (Table 3).

The loading efficiency term interpolates between a floor λ_g and unity. It equals one at night and $\lambda_g + (1-\lambda_g)(1-k_s)$ in the light, so export is least efficient when the light-phase partition is strongly biased to starch. The remaining costs are

$$C_u = c_N\, U_N + c_P\, U_P, \tag{30}$$

$$C_t = \tau_c\left(r_m\, s + \eta\, E\right), \tag{31}$$

$$R_m = r_m\, s, \tag{32}$$

$$C_g = \begin{cases} Y\left(\dfrac{dB_l}{dt} + \dfrac{dB_r}{dt}\right), & \dfrac{dB_l}{dt} + \dfrac{dB_r}{dt} > 0, \\ 0, & \text{otherwise.} \end{cases} \tag{33}$$

The uptake cost C_u prices the realised nutrient fluxes at the sucrose consumption coefficients c_N and c_P, and is evaluated after Eq. (35) so that it uses the fluxes of the current state. The transport cost C_t is a fixed fraction τ_c of the respiratory and loading fluxes. Maintenance respiration R_m is first order in the cytosolic pool with specific rate r_m, charged over all hours as in the source formulation [55]. The construction cost C_g prices the biomass actually built, at Y moles of hexose per gram of fresh biomass. It is applied after both growth rates have been evaluated and is one-sided: senescence returns tissue but does not refund construction carbon.

The term is present in the formulation but inactive in the results. Every run reported in this paper uses $Y = 0$, which is the value carried by both the shipped configuration and the compiled default, so no carbon is charged for the biomass that is built. The growth law of Eq. (42) therefore converts sucrose to biomass without paying a stoichiometric price, and the coefficient λ_{sb} absorbs that conversion empirically. Two consequences follow and we state them here rather than leave them implicit. The carbon budget of Section 4.3 is a bookkeeping statement about the carbohydrate pools and not a whole-plant carbon balance, and the growth coefficient calibrated in Section 6.1 is a rate constant, not a measured yield. The term is dormant and outside the formulation that this paper evaluates, and activating it is not a matter of substituting a literature value. Eq. (33) multiplies Y, in $\mu\text{mol C}_6\,\text{gFW}^{-1}$, by a biomass rate in gFW h^{-1}, giving $\mu\text{mol C}_6\,\text{h}^{-1}$, while every other term of the sucrose equation is written on the nominal per-fresh-weight basis of Section 3.2. The two bases would have to be reconciled before a sourced Y could be switched on, and the reconciliation would change the calibration. We record the mismatch here rather than present activation as a substitution.

3.9. Submodel S5b: nutrient uptake and costs

Nutrient acquisition is active and therefore carbon-limited (A5). It is represented in three stages: a soil-availability rate, an allocation of root capacity, and a sucrose-limited realisation.

The soil stage is Michaelis–Menten in the external concentration [14, 58],

$$I_N = I_{N,\max}\,\frac{N_s}{N_s + K_N}, \qquad I_P = I_{P,\max}\,\frac{P_s}{P_s + K_P}. \tag{34}$$

The potential uptake scales this by the investment state and by the root fraction of total biomass, and the realised uptake applies a sucrose limitation:

$$\widehat{U}_N = I_N\,\alpha_N\,\frac{B_r}{B+\varepsilon}, \qquad U_N = \widehat{U}_N\,\frac{s}{s + c_N\,\widehat{U}_N\,\tau + \varepsilon}, \tag{35}$$

$$\widehat{U}_P = I_P\,\alpha_P\,\frac{B_r}{B+\varepsilon}, \qquad U_P = \widehat{U}_P\,\frac{s}{s + c_P\,\widehat{U}_P\,\tau + \varepsilon}.$$

The saturating factor in Eq. (35) throttles uptake by carbon status. The quantity $c_N\,\widehat{U}_N\,\tau$ is the sucrose that potential uptake would consume over a reference interval $\tau = 1$ h, so it is comparable with s: the factor approaches one when the pool is ample and falls when carbon is short. The constant ε guards the denominators only and is not a model parameter.

The internal pools are balanced against two demands, structural and photosynthetic:

$$\frac{dN}{dt} = U_N - \left[r_m\,(s+a) + \eta\,E\right] r_{A,N}\,\frac{B_l}{B+\varepsilon} - \frac{\Phi_c}{\Phi_{c,\max}}\,N_{ph}\,\nu, \tag{36}$$

$$\frac{dP}{dt} = U_P - \left[r_m\,(s+a) + \eta\,E\right] r_{A,P}\,\frac{B_l}{B+\varepsilon} - \frac{\Phi_c}{\Phi_{c,\max}}\,P_{ph}\,\nu. \tag{37}$$

The first loss term charges nutrients against the carbon turnover of the shoot, at the assimilation cost coefficients $r_{A,N}$ and $r_{A,P}$ and in proportion to the leaf share of biomass. The second charges the nutrient requirement of photosynthesis, scaled by the relative assimilation rate $\Phi_c/\Phi_{c,\max}$ and the conversion parameter ν, with N_{ph} and P_{ph} the contents below which assimilation would be nutrient limited. Those thresholds appear in the saturation functions

$$\varphi_N = \max\!\left(0,\ \frac{2N}{N+N_{ph}} - 1\right), \qquad \varphi_P = \max\!\left(0,\ \frac{2P}{P+P_{ph}} - 1\right), \tag{38}$$

which are zero at the threshold and approach one at large content.

These two functions are the only place where nutrient status could reach carbon acquisition, and in the configuration reported here they do not reach it. They enter the prescribed-input variant M0 of Section 6.4 through a multiplicative factor $\min(\varphi_N, \varphi_P)$ on the assimilation rate. The Farquhar routine used for every reported result contains no nutrient term, so under M1 and M2 assimilation is independent of N and P, and the coupling from nutrients to growth runs through the sucrose economy alone. The architecture of Figure 1 shows no arrow from the nutrient states to the gas-exchange layer for this reason.

The investment states α_N and α_P carry the acclimation of uptake capacity. Each follows a two-term law. The first term drives investment up when the nutrient is scarce or assimilation is high. The second drives it down when the pool is full, or when the carbon price of uptake is high relative to the available sucrose flux:

$$\frac{d\alpha_N}{dt} = (1-\alpha_N)\left[\left(1 - \frac{U_N}{U_N + C_N + \varepsilon}\right)\frac{N_{\max}}{N+N_{\max}} + \frac{\Phi_c}{\Phi_{c,\max}} - \Omega\right] - \alpha_N\left[\frac{N}{N+N_{\min}} + \frac{c_N U_N}{c_N U_N + \Phi_c\,(1-k_s) + D_a + \varepsilon}\right], \tag{39}$$

$$\frac{d\alpha_P}{dt} = (1-\alpha_P)\left[\left(1 - \frac{U_P}{U_P + C_P + \varepsilon}\right)\frac{P_{\max}}{P+P_{\max}} + \frac{\Phi_c}{\Phi_{c,\max}} + \Omega\right] - \alpha_P\left[\frac{P}{P+P_{\min}} + \frac{c_P U_P}{c_P U_P + \Phi_c\,(1-k_s) + D_a + \varepsilon}\right], \tag{40}$$

where C_N and C_P are the nutrient cost terms and Ω is the stoichiometric imbalance signal

$$\Omega = \alpha\, \lambda_k\, \frac{N - \omega P}{N + \omega P + \varepsilon}, \tag{41}$$

with α the corresponding investment state, ω the optimal N:P ratio and λ_k the gain. The sign of Ω is opposite in the two equations, which is the coupling that makes the pair competitive: an excess of nitrogen relative to the optimal ratio suppresses further nitrogen investment and promotes phosphorus investment. The factors $(1-\alpha)$ and α again keep each state in $[0, 1]$ without a clamp.

3.10. Submodel S6: biomass growth and allocation

Growth is driven by the phloem-loading flux and controlled by the allocation state ρ_r, the fraction of that flux directed to the root. Both organs lose biomass to mortality and to a quadratic self-competition term:

$$\frac{d\,B_l}{dt} = \lambda_{sb}\left(1-\rho_r\right)\eta\, E\, B_l \;-\; \mu_l B_l \;-\; \gamma_l B_l^2, \tag{42}$$

$$\frac{d\,B_r}{dt} = \lambda_{sb}\,\rho_r\,\eta\, E\, B_l \;-\; \mu_r B_r \;-\; \gamma_r B_r^2. \tag{43}$$

Two features of Eqs. (42)–(43) are easy to misread. Both growth terms are proportional to *leaf* biomass B_l and not to the biomass of the organ being built, because the leaf is the source of the loading flux; root growth is therefore driven by the size of the source. The loading flux also appears in both equations with the same coefficient λ_{sb}, so the split between organs is carried entirely by ρ_r. That coefficient converts the product of loading flux and leaf biomass into a growth rate. The specific loading flux ηE has unit μmol C_6 gFW^{-1} h^{-1}, so the product $\eta E B_l$ has unit μmol C_6 h^{-1} and λ_{sb} must carry gFW (μmol C_6)$^{-1}$ for $d\,B_l/dt$ to come out in gFW h^{-1}. It is a calibrated coefficient and not a construction stoichiometry; the construction stoichiometry is Y in Eq. (33). The quadratic terms bound the biomass trajectory at large size and are negligible at the seedling scale of the reported runs.

The allocation state responds to the balance between the nutrient investment already made and the nutrient and carbon status achieved:

$$\frac{d\rho_r}{dt} = \underbrace{\left(1-\rho_r\right)\left[\alpha_N\varsigma + (1-\varsigma)\,\alpha_P\right]}_{\text{demand for soil resources}} \;-\; \underbrace{\rho_r\left[\frac{N\,\varsigma}{N+N_{\min}} + \frac{P\,(1-\varsigma)}{P+P_{\min}} + \frac{s_{\min}}{s_{\min}+s}\right]}_{\text{satisfied demand}}, \tag{44}$$

where the stoichiometric weight

$$\varsigma = \frac{\omega}{\omega + \max(N,0)/\left(\max(P,0)+\varepsilon\right)} \tag{45}$$

shifts attention between the two nutrients according to which is further from the optimal ratio ω. The positive part in Eq. (45) makes the ratio well defined for a numerically negative pool and is the identity wherever $N \geq 0$ and $P \geq 0$. Equation (44) is the functional-balance statement of the model [56, 57]: investment moves to the root while soil resources are in demand, and away from it once nutrient and carbon status are satisfied. Both terms carry the factor ρ_r or $(1-\rho_r)$ and keep the state in $[0, 1]$.

Soil nutrient status enters the growth coefficient through a multiplicative gate. With scaled soil concentrations $n = N_s/N_s^{\mathrm{ref}}$ and $p = P_s/P_s^{\mathrm{ref}}$,

$$\lambda_{sb}^{\mathrm{eff}} = \begin{cases} \lambda_{sb}, & \min(n,p) \geq 1, \\ \lambda_{sb}\,\max\!\left(0,\; (n+p) - \frac{1}{2}\left(n^2+p^2\right)\right), & \text{otherwise.} \end{cases} \tag{46}$$

The gate is inactive at or above the reference concentrations and reduces the growth coefficient below them. In every scenario reported here the soil concentrations are at or above those values, so $\lambda_{sb}^{\mathrm{eff}} = \lambda_{sb}$ throughout. We give the gate because it is part of the right-hand side and because the reference concentrations are hard-coded divisors rather than configurable parameters, as Table 2 records.

3.11. Bounded constitutive laws

The right-hand side contains fifteen saturating ratios of the form $X/(X + K)$ with $K > 0$, each with a pole at $X = -K$. Those poles lie outside the physically meaningful region but not outside the region an adaptive integrator can probe. A trial stage that takes a nearly depleted pool slightly negative can land close to a pole and return a derivative many orders of magnitude too large, after which the step controller cannot meet its error estimate at any step size. The formulation therefore evaluates every constitutive law at the positive part of each pool state,

$$\widetilde{X} = \max(X, 0), \qquad X \in \{a,\ s,\ s_v,\ N,\ P,\ B_l,\ B_r\}, \tag{47}$$

and uses $\widetilde{X}$ wherever a pool appears inside a saturating ratio or a cost term. The state vector is not modified and the differential equations are unchanged. Equation (47) is the identity on $X \geq 0$, the whole region in which a pool content is defined, so it cannot alter a physically admissible trajectory. It states where the constitutive laws are defined and is not an additional model assumption. The states k_s, α_N, α_P and ρ_r need no such treatment, since their equations are self-bounding through the $(1 - \cdot)$ factors above.

3.12. Parameters and their provenance

Every parameter carries a provenance record in the released repository, in a machine-readable ledger that names its value, unit, source and evidence tier. The six tiers are T1 measured, T2 derived, T3 inherited from the source model, T4 constrained by physics or chemistry, T5 fitted, and T6 unsourced. A value may not be defended by the improvement it produces in the fit; its justification rests on the tier, not on the residual. Table 2 is the curated physical parameter table generated from that ledger, and the complete ledger with per-entry sources is reproduced in Appendix B.

Of the 64 model parameters, 30 are literature-sourced or derived (2 at T1, 8 at T2, 20 at T3), 7 are constrained by physical argument (T4), 15 are fitted (T5) and 12 remain unsourced (T6). We report the unsourced count rather than distribute it, because it states what is not yet established. The T6 entries are of three kinds. The first is values in common use that we could not trace to a primary measurement in a comparable system, such as the compensation point Γ^* and the quantum yield ϕ. The second is constants hard-coded rather than exposed in the configuration file, including several Jarvis response coefficients. The third is one structurally questionable value, the starch capacity, whose ledger row says so. Appendix B gives each entry. Of the 15 fitted entries, 11 are inherited fitted values of the source model that are not re-estimated here.

The remaining four fitted parameters are given separately in Table 3, because the number of free parameters is the central claim of the evaluation. All four are shared: one value of each is used at every photoperiod in every reported run. The legacy per-photoperiod baseline instead calibrated eight parameters, two coefficients at each of four photoperiods, against eight observations. That leaves zero degrees of freedom and makes its reported agreement a calibration residual rather than a prediction. Identifiability of the shared set is examined in Sections 5 and 6.

Two ledger entries affect every run and need a note here. The maximal conductance $g_{s,\max} = 1/\sqrt{2}$ is derived, not chosen: it makes the corrected normalisation of Eq. (11) an exact reparametrisation at $T_0 = 25\,°\mathrm{C}$, so the product $g_{s,\max} f_T(T_0)$ is unchanged and no downstream parameter needs recalibration. The construction stoichiometry Y, taken by the source formulation from the growth-respiration yields of Penning de Vries et al. [60], is set to zero in the released configuration, so $C_g = 0$ in Eq. (27) in every run reported here; we report the zero as a stated simplification rather than carry a derived non-zero value the code does not use, and Section 4.3 states the consequences. The specific leaf area is the one T1 measured value in the gas-exchange block [59], and the vacuolar release rate k_{out} is derived from a measured vacuolar-to-cytosolic sucrose ratio [47]; both derivations are set out in Appendix B.

The full parameter table with all provenance fields is given in Appendix B; Tables 2 and 3 give the parameters that appear in the equations of this section.

4. Numerical Implementation

The model is implemented as a modular simulation engine in C (ISO C99) and built with CMake. Environmental forcing, gas exchange, carbohydrate dynamics, nutrient uptake and biomass growth are written as separate translation units. Each physiological block can therefore be tested on its own, while the coupled path from forcing to growth stays intact. This section describes how the state equations are advanced, and which checks were applied to the solver components before any result was produced.

4.1. Simulation architecture and time integration

Each right-hand-side evaluation follows the fast–slow structure of Section 3 and Figure 1. Four blocks are executed in sequence. The first block evaluates the clock time and the environmental drivers: PAR, leaf temperature, relative humidity, vapour-pressure deficit, ambient CO_2 and leaf water potential. The second block computes the gas-exchange layer: stomatal conductance, the electron transport rate and the two Farquhar branches. The intercellular CO_2 concentration follows from the closed-form solution of Eq. (20). The third block converts net assimilation into the carbon-input term Φ_c. The fourth block assembles the derivatives of the eleven slow states. These are starch, cytosolic and vacuolar sucrose, the partition and allocation variables, the two nutrient pools, the two affinity states, and leaf and root biomass.

Gas exchange is algebraic and is solved at every stage of every step, not once per output sample. This matters for the coupling. Conductance, assimilation and C_i are functions of the current state and the current forcing, so a stage value taken from a stale operating point would break the supply–demand balance that the closure enforces. The eleven slow states carry no algebraic constraint and are integrated as an explicit system.

Integration uses a separate, reusable C ODE-solver library through a single interface, so the model does not depend on one backend. The production configuration selects an adaptive Dormand–Prince embedded pair (`ODE45`) with an error tolerance of 10^{-8} [21]. Under an adaptive method the configuration entry `dt_hours` is not the integration step. It sets the initial trial step, which the step controller immediately overrides, and it sets the interval at which the trajectory is sampled for output. Production runs use `dt_hours` $= 0.1$ h, so the reported trajectory is a fixed 0.1 h grid over a 30-day horizon, while the accuracy of that trajectory is controlled by the tolerance alone.

A fixed step is used where the adaptive controller is either unusable or unnecessary. The sensitivity and sweep runners override the backend to fixed-step RK4, because those designs draw parameter values at the edges of the declared ranges, where the adaptive step can collapse. A fixed step also gives a predictable cost over several thousand evaluations. The carbon-balance tool of Section 4.3 uses fixed-step RK4 for a different reason: only a fixed-step driver can advance the flux accumulators with the same weights and the same step as the states. Both backends are run through the same production path at the same settings, so the effect of the choice can be measured rather than assumed.

Two runtime guards protect batch work. A non-finite guard and a domain guard observe the trajectory and classify a run as divergent. A step-budget plugin counts attempted steps and stops the run when a generous budget is exhausted. The second guard covers a failure the first cannot see: for some parameter combinations the adaptive step collapses while the state stays finite, so the integrator neither advances nor reports an error. Such a run then unwinds normally and is recorded as inadmissible, which is how the batch drivers already treat divergence.

4.2. Verification of the solver components

The gas-exchange closure was verified by substitution rather than by assertion. For a reference leaf and PAR from 10 to 2000 μmol m^{-2} s^{-1}, the analytic solution was inserted back into the supply relation and the residual $|A_n - g_c(C_a - C_i)|$ was recorded. The analytic branch solution reproduces the intended operating point. The residual stays at machine precision over the whole light range, so the closed-form root solves the coupled system rather than approximating it.

Two limits were checked separately, because both are boundaries where an implementation can silently disagree with the equations. When conductance reaches zero the implementation returns $C_i = \Gamma^*$ and $A_n = -R_d$, which is the dark limit of the formulation. Above that limit, assimilation crosses zero smoothly at PAR $= 13.36$ μmol m^{-2} s^{-1} under the reference conditions. An earlier hard cut-off at PAR $= 20$ μmol m^{-2} s^{-1} was removed, because f_{PAR} is not zero there and the returned assimilation jumped across the threshold. With the analytic closure the terminal $\max(0, \cdot)$ clamp already returns exactly zero below the light compensation point, continuously and without a special case. Removing the gate therefore also removes a discontinuity that an adaptive step controller has to resolve twice per simulated day.

4.3. Carbohydrate-subsystem bookkeeping closure

Whether the carbon bookkeeping closes cannot be answered by post-processing the solver output. The trajectory records pool sizes, whereas closure is a statement about time-integrated fluxes, and a quadrature applied afterwards would test the quadrature as much as the model. The carbon-balance tool therefore appends ten flux accumulators to the eleven states and advances the augmented system with identical RK4 weights and an identical step. Both parts come from one call to the right-hand side, so the accumulators see the same stage values as the state integration. The resulting budget is then exact to the arithmetic of the integrator.

The closure is asserted for the carbohydrate subsystem $\{a, s, s_v\}$, that is starch, cytosolic sucrose and vacuolar sucrose. Adding Eqs. (25), (27) and (28) makes the partition terms $k_s\Phi_c$ and $(1 - k_s)\Phi_c$ recombine, the degradation

flux D_a cancel between the starch and sucrose equations, and the vacuolar exchange $k_{\mathrm{in}} s - k_{\mathrm{out}} s_v$ cancel between the sucrose and vacuolar equations. Integrating what remains over $[0, T]$ gives

$$\left(a + s + s_v\right)\Big|_0^T = \int_0^T \Phi_c \, dt - \int_0^T \left(C_u + C_t + R_m + \eta \, E + C_g\right) dt, \tag{48}$$

with every symbol as defined in Eqs. (30)–(33) and (27), so the identity can be read off the model equations without translation. Two points about the terms. The integrand on the input side is the model carbon input Φ_c of Eq. (22), in μmol hexose gFW^{-1} h^{-1}, and not the leaf-level A_n: the accumulator integrates the same converted quantity that drives the pools, so both sides carry one set of units. And the construction cost C_g is identically zero in every reported run because $Y = 0$, so the implementation carries no accumulator for it: the ten accumulators are assimilation, the two partition fluxes, starch degradation, the uptake, transport, maintenance and loading costs, and the two vacuolar exchange fluxes. The term is written in the identity because it is required whenever Y is activated, and a reader checking the residual against the code should expect it to be absent there for that reason. The residual is the difference between the two sides, reported relative to cumulative carbon input,

$$\varepsilon_C = \frac{1}{\int_0^T \Phi_c \, dt} \left[\left(a + s + s_v\right)\Big|_0^T - \int_0^T \Phi_c \, dt + \int_0^T \left(C_u + C_t + R_m + \eta \, E + C_g\right) dt \right]. \tag{49}$$

Across the four calibration photoperiods the relative residual lies between 2.9×10^{-14} and 6.1×10^{-14}, which is machine precision for double-precision accumulation over this many steps. No carbon is created or destroyed numerically inside the declared scope.

The scope must be stated exactly, because it does not cover the whole plant. The biomass gain term sits outside the budget by construction. In the growth law of Eq. (42) the sucrose drain is $\phi_{\mathrm{load}} E_{\mathrm{night}}$ while the biomass gain is $\lambda_{sb} \, \phi_{\mathrm{load}} E_{\mathrm{night}} B_l$, so the two differ by the factor $\lambda_{sb} B_l$. That factor equals unity only at $B_l = 1/\lambda_{sb}$, which is of order 10^2 gFW. Over the simulated seedling range the factor stays 10^2 to 10^3 away from unity, so the two terms are not two views of one carbon flow. The growth term is a rate law calibrated against observed RGR, not a stoichiometric conversion of sucrose into biomass carbon. Eq. (49) therefore verifies bookkeeping closure of the carbohydrate subsystem, and it is reported under that name only. A budget that included biomass would require a stoichiometric growth equation, which the model does not have.

4.4. Convergence and tolerance study

Three numerical settings can in principle contaminate a reported number: the solver tolerance, the output interval and the gas-exchange tolerance. Each was refined independently around its production value, together with a step sweep of the fixed-step backend and a comparison of the two backends at production settings. Every step used divides both 24 h and the photoperiod exactly, so the end-of-day sample lands on a grid point and no interpolation error enters.

The reported quantities are insensitive to these settings over the tested range. At the production step of 0.1 h the worst-case relative deviation from the $dt \to 0$ reference over the four photoperiods is 4.3×10^{-4} in RGR, 3.1×10^{-3} in final biomass and 2.0×10^{-3} in end-of-day starch; the cytosolic sucrose pool is the least well resolved at 6.8×10^{-2}, as the smallest and fastest-turning state. The algebraic tolerance inside the C_i solve moves RGR by at most 3.2×10^{-7}, and the production configuration uses the closed-form solver, which has no such tolerance. The measured order of the fixed-step sweep is close to one rather than four, because starch degradation is multiplied by the light state and the right-hand side is discontinuous twice per simulated day. None of these deviations approaches the 10^{-2} to 10^{-1} discrepancies the model shows against measurement, so numerical error is not what limits agreement. The full study, including the Richardson estimates, the per-photoperiod tables, the legacy iterative comparator and the one external integrator excluded after it failed a consistency check, is in `docs/revision/convergence_study.md`.

5. Model Evaluation Protocol

This section defines the evaluation rules and reports no results. The reference data are stated with their source, the fitted coefficients are counted in each regime, and every metric is defined together with what it does not measure. No number in Section 6 should then be readable as more than it is.

5.1. Reference data and growth conditions

Four photoperiods are used for calibration and cross-validation: 4, 6, 8 and 12 h of light per 24 h cycle. An 18 h photoperiod is held out of every fit entirely and is used only as an out-of-sample condition. The four short photoperiods span the range over which the reference experiments report growth, and 18 h lies outside it, which is why it is treated separately.

All reported runs reproduce the growth conditions of the reference experiments. PAR is 160 μmol m^{-2} s^{-1} delivered as a square wave over the light phase, and leaf temperature is 21 °C by day and 19 °C by night. This is a methodological requirement, not a detail. The shipped configuration instead runs a half-sine profile peaking at 1200 μmol m^{-2} s^{-1} at a single temperature. A half-sine averages $2/\pi$ of its peak, so that daily light integral is about 4.8 times the chamber value at the same day length. A coefficient fitted under one light dose does not transfer to another: the carbon-input coefficient would absorb the dose mismatch, and the resulting value would describe the forcing rather than the plant. Scoring the model against chamber data under the shipped forcing would therefore compare a model driven at one photon dose with plants grown at another.

Every reference value used in this paper comes from one study [54], and we state how that was established because an earlier draft of this work attributed part of the data to the wrong source. The RGR targets are 0.068, 0.114, 0.171 and 0.260 d^{-1} for the 4, 6, 8 and 12 h photoperiods. The diel carbohydrate reference is `data/paper/sugars_starch_reference.csv`, which holds 65 rows over five photoperiods, that is 13 sampling times per photoperiod at a 2 h resolution through the 24 h cycle, with the row at ZT = 24 a copy of ZT = 0. Starch and sucrose are used; the remaining assayed metabolites are not part of this evaluation.

The diel series were traced to Supplemental Table S4 of that study, the sheet of replicate means, and independently verified against the published supplement by a comparison script released with the code, so the identification can be repeated rather than taken on trust.

That the growth rates, the dusk starch values and the diel series all come from one growth experiment is what makes the reference set internally consistent: a model scored against it is not being asked to reconcile two laboratories. Two other studies of diel carbohydrate behaviour cover only a subset of these day lengths, at 12 h [1] and at 6 and 12 h [36], and neither can supply the four-point series.

5.2. Parameterisation regimes

Three parameterisation regimes are evaluated (Table 4). They differ only in how many coefficients are fitted and in whether those coefficients are allowed to depend on photoperiod. Everything else, including the forcing, the initial state and all sourced parameters, is identical.

The saturated regime is the legacy per-photoperiod baseline. It fits two coefficients per photoperiod by nested bisection: the growth coefficient λ_{sb} against the observed RGR, and the daytime starch synthesis rate λ_{sdi} against the observed end-of-day starch. With four photoperiods that is eight fitted values against eight observations, so the regime has zero degrees of freedom. Its agreement with the data cannot be reported as predictive accuracy. What a bisection leaves behind is a convergence residual of the search, controlled by the resolution of the parameter grid, and it goes to zero as the search is tightened. The regime is retained so that the published result stays reproducible, and it is reported as a calibration diagnostic only.

The shared regime fits four coefficients for all photoperiods together: λ_{sb}, λ_{sdr}, λ_{sdi} and the sink activation constant. Photoperiod is then the only thing that differs between the four runs, so any photoperiod response the model produces is generated by its dynamics rather than supplied by its parameters. The third regime keeps the same shared set but allows the growth coefficient to vary linearly with photoperiod P,

$$\lambda_{sb}(P) = c_0 + c_1 P, \tag{50}$$

which replaces one shared constant by two coefficients. Eq. (50) encodes the hypothesis that the growth response to photoperiod is smooth in P and is not fully generated by the coupled dynamics. It remains a prediction for a held-out photoperiod, because the held-out observation never enters the fit.

5.3. Metrics

Growth is evaluated on relative growth rate, computed from total biomass over the simulated interval,

$$\mathrm{RGR} = \frac{\ln B(t_2) - \ln B(t_1)}{t_2 - t_1}. \tag{51}$$

Per photoperiod we report the signed relative error. Over a set of photoperiods we report the mean absolute percentage error:

$$E_{\mathrm{rel}} = 100\,\frac{\widehat{\mathrm{RGR}} - \mathrm{RGR}_{\mathrm{ref}}}{\mathrm{RGR}_{\mathrm{ref}}}, \qquad \mathrm{MAPE} = \frac{1}{m}\sum_{j=1}^{m}\left|E_{\mathrm{rel},j}\right|, \tag{52}$$

where $\widehat{\mathrm{RGR}}$ is the simulated value and m is the number of photoperiods in the set. In-sample and held-out MAPE are always reported as separate quantities and never pooled.

The diel carbohydrate series are evaluated with a normalised root-mean-square error. A single multiplicative scale α is first fitted to each simulated trace by least squares,

$$\alpha = \frac{\sum_{i=1}^{n}\hat{y}_i y_i}{\sum_{i=1}^{n}\hat{y}_i^2}, \qquad r_i = \alpha\hat{y}_i - y_i, \tag{53}$$

and the metrics are then computed on the scaled residual r_i,

$$\mathrm{nRMSE} = \frac{1}{|\bar{y}|}\sqrt{\frac{1}{n}\sum_{i=1}^{n} r_i^2}, \qquad \mathrm{MAE} = \frac{1}{n}\sum_{i=1}^{n}|r_i|. \tag{54}$$

Here y_i is the reference value and $\hat{y}_i$ the simulated value, n is the number of matched sampling times, and $\bar{y}$ is the mean reference level. MAE keeps the units of the reference data and is computed for the same fitted α, so both metrics describe one and the same fit.

The consequence of fitting α must be stated, because it bounds what the carbohydrate comparison can support. A fitted multiplicative scale removes the absolute level from the residual, so nRMSE scores the shape and the timing of the diel cycle and not the absolute pool concentration. A trace with the correct rhythm and a systematically wrong amplitude can pass; a trace with the correct amplitude and the wrong phase cannot. For that reason α is reported beside every nRMSE in Section 6: a value near unity means shape and level agree, and a value far from unity localises the disagreement in the level alone. The acceptance thresholds are predefined heuristic criteria rather than external standards: nRMSE ≤ 0.25 for starch and nRMSE ≤ 0.50 for sucrose. The sucrose threshold is the looser of the two because whole-tissue sucrose pools cytosolic and vacuolar sucrose, and the split between the two compartments is itself uncertain.

5.4. Cross-validation design

Growth prediction is evaluated by leave-one-photoperiod-out cross-validation. The split is by photoperiod rather than at random, because interpolation across the photoperiod range is exactly the ability in question. Four folds are run in each of the two shared regimes, which gives eight held-out predictions in total.

One coefficient is refitted per fold, and stating this precisely bounds what the experiment can support. The growth coefficient λ_{sb} is searched on a fixed grid of eleven values from 0.00200 to 0.00480 gFW $(\mu\mathrm{mol\ C_6})^{-1}$. For each grid value the model is run at all four photoperiods, and the value minimising the mean absolute percentage error in RGR over the three training photoperiods is selected, frozen, and used to predict the fourth. In regime A the selected value is applied unchanged at the held-out photoperiod. In regime B the straight line of Eq. (50) is fitted to the three per-photoperiod grid optima of the training set by least squares, and evaluated at the held-out photoperiod. Regime B therefore carries two coefficients where regime A carries one.

The other three calibrated coefficients, λ_{sdr}, λ_{sdi} and K_s, are not refitted in any fold: they keep the values obtained from the full four-photoperiod calibration, as do all sourced parameters. The design measures how much of the photoperiod response is carried by the growth coefficient specifically, with the carbohydrate coefficients held fixed. It does not establish that this coefficient is the only one that could carry that response, and a design refitting all four per fold could distribute the same dependence differently. A grid search is used rather than a local optimiser because the objective is one-dimensional and bounded, so the grid resolution is the only accuracy limit and no starting point or convergence criterion enters the result.

The 18 h condition is not part of any fold. It is held out of every fit in every regime, and it is evaluated only after the coefficients have been frozen. It therefore tests extrapolation beyond the calibrated range, which is a different and stronger question than the interpolation the four folds test.

5.5. Ablation protocol

The ablation measures what the computed carbon input contributes, by replacing the carbon-input pathway and leaving everything downstream untouched. Three variants are compared. In M0 the carbon input is prescribed by the simple saturating photosynthesis expression of the reference formulation. In M1 the carbon input comes from Farquhar C3 photosynthesis, with the supply–demand closure solved as in the full model, but with stomatal conductance held fixed. In M2 conductance is computed from the Jarvis multiplicative response, so the gas-exchange layer becomes sensitive to PAR, vapour-pressure deficit, temperature, ambient CO_2 and leaf water potential.

All parameters are shared and frozen across the three variants. This is the point of the design and not a simplification. If each variant were re-fitted against the same observations, the fit would absorb the difference between the carbon-input pathways. The comparison would then measure how well a per-variant calibration can compensate, rather than what removing a mechanism costs. The comparison is reported on RGR error per photoperiod. RGR alone cannot separate M1 from M2, so we also report what the coupling changes. Two quantities carry that: the magnitude and diel shape of the carbon input, and the response of the gas-exchange layer to the environmental drivers.

5.6. Sensitivity analysis protocol

The global sensitivity analysis covers the 16 parameters listed in `config/sensitivity_params.csv`. The admissible range, the sampling scale and the literature basis of each are given individually in Appendix C. The set spans nutrient uptake capacity and soil supply, gas-exchange capacity, the environmental drivers, respiration, the growth coefficient, vacuolar exchange and the initial carbohydrate and biomass state. The analysed output is RGR. Every reported index is conditional on these declared ranges: the analysis describes model behaviour over the declared domain and not parameter importance in general.

Morris screening follows Morris [39] with the elementary-effect measures μ^* and σ of Campolongo et al. [8]. The design uses 30 trajectories on a 6-level grid with the fixed step $\Delta = p/[2(p-1)]$, giving $r(d+1)$ model evaluations. Trajectory base points are drawn from an independent pseudo-random generator, as the method specifies, so the screening is not affected by the base-sample construction described below. Variance-based indices use the estimators of Sobol' [52] in the form recommended by Saltelli et al. [44], with $n = 512$ base samples and hence $n(d+2)$ evaluations over the $d = 16$ parameters. Confidence intervals come from 1000 bootstrap replicates of the model evaluations, resampled by base row so that each resampled design remains a valid Saltelli design.

The base-sample design is stated explicitly, because the estimator depends on it. The Saltelli estimator needs two independent base matrices $\mathbf{A}$ and $\mathbf{B}$ of size $n \times d$, and it attributes variance to a parameter through a single-column substitution between them. Any correlation between the corresponding columns therefore biases the index towards zero. The construction specified by Saltelli et al. [44] draws one low-discrepancy sequence in $2d$ dimensions and splits it by column, but the sampler used here supports at most 16 dimensions and $2d = 32$ at $d = 16$. The two matrices are therefore drawn as two independent Latin-hypercube samples, stratified per dimension and independent by construction [34]. The design actually used is written to a `base_design` column of every output file, so no reported index is ambiguous about its own provenance. The estimator implementation was checked against the Ishigami function, whose indices are known in closed form [23].

Uncertainty is reported at two levels, because the two levels answer different questions. Bootstrap intervals resample within one design, so they measure sampling noise given that design and are blind to error contributed by the design itself. Across-design spread was therefore checked separately, by repeating the whole analysis on independently generated base samples over base sample sizes from $n = 512$ to $n = 4096$. An estimator whose across-design spread does not shrink as n grows is misspecified rather than noisy, so both levels are needed before an index is interpreted. Divergent evaluations are counted and reported with the indices, and no evaluation is silently discarded, because deleting rows from a Saltelli design breaks the pairing the estimator relies on. One-at-a-time sweeps and two-parameter response surfaces are also computed, but they are local diagnostics of the response shape and are not used as a global ranking.

6. Results

All results below use one code version and one configuration file. Unless a subsection says otherwise, the forcing is the chamber condition of the reference experiment and the simulated period is 30 days. The shared parameterisation has four fitted coefficients: λ_{sb}, λ_{sdr}, λ_{sdi} and the half-activation sucrose concentration of the sink term. These values are the same at every photoperiod and are not refitted between the studies below.

6.1. Photoperiod-dependent relative growth rate

With the shared parameterisation the model gives a mean absolute percentage error of 8.58% in relative growth rate (RGR) across the four photoperiods, and the worst photoperiod reaches 11.75%. All four scenarios integrate to the end of the simulated period without a non-finite sample. Three of the four photoperiods sit between 10.1 and 11.8% and the 6 h scenario is substantially better at 1.28%; we report the continuous error rather than a pass or fail verdict, because any threshold we could name here would be one we had chosen ourselves. Table 5 lists the per-photoperiod values and Figure 2 shows them against the reference targets of Sulpice et al. [54].

The errors are not evenly distributed, and their signs matter. The 4 h scenario is overestimated by 11.21% and the 6 h scenario by 1.28%, while 8 and 12 h are underestimated by 10.10 and 11.75%. The model therefore rotates the photoperiod response: it is too steep at the short end and too shallow at the long end. One parameter set reproduces the magnitude and the direction of the response, but not its exact slope. Section 6.6 shows that this residual has a structural cause and not a calibration cause.

The growth response is large in absolute terms, which is worth stating because RGR is a logarithmic rate and hides it. Final total biomass after 30 days rises from 0.387 gFW at 4 h to 1.258 at 6 h, 4.006 at 8 h and 39.03 gFW at 12 h, a hundredfold spread produced by a threefold change in day length. The errors quoted above are therefore errors in an exponent, and a 10% error in RGR compounds over the simulated period. Halving the output step changes the simulated RGR and the final biomass by less than 0.01% in all four scenarios, so these differences come from the model and not from the integration.

The saturated per-photoperiod regime behaves differently, and the contrast matters. That regime fits two coefficients per photoperiod, eight against eight observations, so it has zero degrees of freedom. Its per-photoperiod errors on the grid used here are 3.42, 1.28, 2.45 and 3.03%. These values are set by the grid resolution of 2.5×10^{-5} rather than by the model, and refining the bisection drives them towards zero. They are convergence residuals and carry no information about predictive skill. The point is not academic. Evaluated under the chamber forcing used throughout this paper, the same per-photoperiod parameterisation gives an RGR error of 48.67% at every photoperiod. Near-zero residuals at the calibration point with a large error one condition away are the expected signature of a saturated fit.

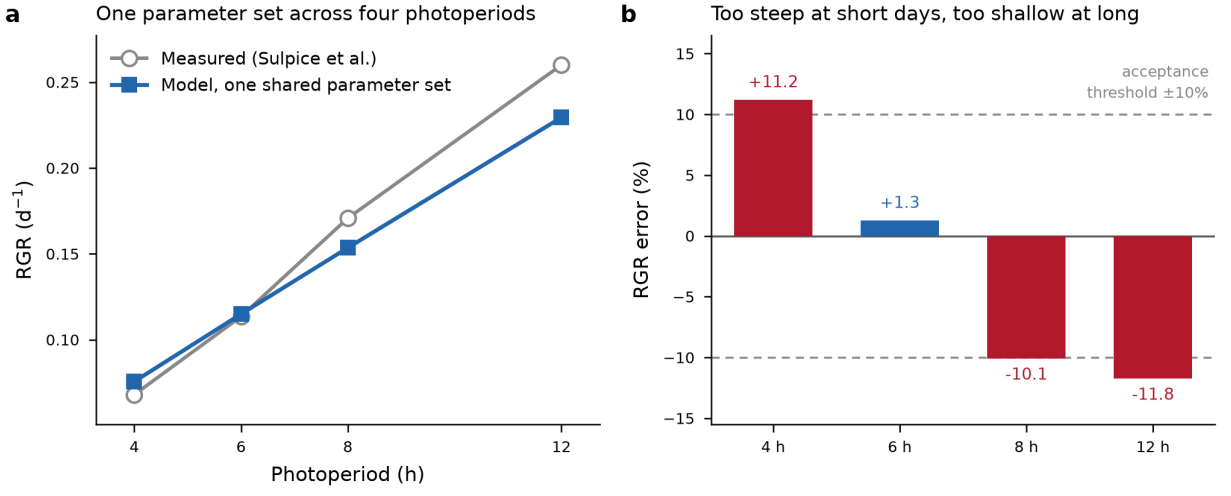


Figure 2: Relative growth rate against photoperiod. Simulated values under the shared four-parameter parameterisation are compared with the reference targets of Sulpice et al. [54] for 4, 6, 8 and 12 h photoperiods. The residuals of the saturated per-photoperiod calibration are shown for reference only; they are convergence residuals of a fit with zero degrees of freedom.

6.2. Cross-photoperiod prediction

The held-out test asks a different question: how well does the model predict a photoperiod that was not used to fit anything? We fitted the growth coefficient on three photoperiods, predicted the fourth, and repeated this for each of the four folds. Table 6 reports both regimes and Figure 3 shows the predictions.

The scope of the refit must be stated, because it bounds what the test can support. Only λ_{sb} is refitted per fold, on a grid of eleven values. The other three calibrated coefficients, λ_{sdr}, λ_{sdi} and K_s, keep the values obtained from the full four-photoperiod calibration in every fold. The experiment therefore measures how much of the photoperiod response is carried by the growth coefficient specifically, holding the carbohydrate coefficients fixed. It does not establish that λ_{sb} is the only coefficient that could carry it, and a design that refitted all four per fold could distribute the same dependence differently.

A separate cross-validation checks whether the result depends on the two coefficients that this design holds at per-photoperiod values. These two are instead fitted as single shared values, with eight deterministic restarts per fold. The qualitative outcome is unchanged. A photoperiod-independent growth coefficient still fails to predict a held-out photoperiod, and a linear dependence still succeeds, with the held-out error changing by less than the spread across folds. The apparent predictive skill is therefore not an artefact of the two frozen coefficients. That study runs under the shipped half-sine forcing rather than the chamber forcing used here, so its error percentages are not directly comparable to those of Table 6; it is reported in `docs/revision/crossval_robust.md`.

In regime A the growth coefficient is one shared constant. The held-out error is 13.18% and the worst fold is 20.93%, with the 4 h treatment held out. In regime B the same coefficient depends linearly on photoperiod P as in Eq. (50), with two coefficients instead of one. The held-out error falls to 4.36% and the worst fold is 10.30%, now at the 12 h treatment. The fitted values are $c_0 = 0.00243$ and $c_1 = 8.786 \times 10^{-5}$ per hour. Across the four folds the fitted growth coefficient rises monotonically with photoperiod, from 0.00275 at 4 h to 0.00370 at 12 h. Adding one coefficient therefore predicts an unseen photoperiod about three times better than a single constant.

Two readings follow. First, the growth coefficient is not photoperiod-invariant in this model. If the coupled dynamics generated the full photoperiod response, a shared constant would be enough and regime B would add nothing. The result is that allowing photoperiod dependence in this coefficient reduces held-out error, which is weaker than saying this coefficient is where the dependence belongs. Second, the missing dependence is compact. A straight line in one coefficient captures it over the tested range, which places it in the conversion from available sucrose to structural growth rather than in the gas-exchange layer upstream. A positive slope means that a longer day converts loaded sucrose into biomass more efficiently, not only for longer. Increased sink activity under long days is a plausible reading [51, 54], but the design cannot separate it from an allocation effect or from a fresh-weight conversion effect.

The limits of Eq. 50 belong with the result. The line was fitted over 4 to 12 h and has no support outside that interval. It corrects a model coefficient and does not describe a process, so extrapolating it to 18 or 24 h has no basis. It must also break down at short photoperiods, because the coefficient would reach zero at a positive day length.

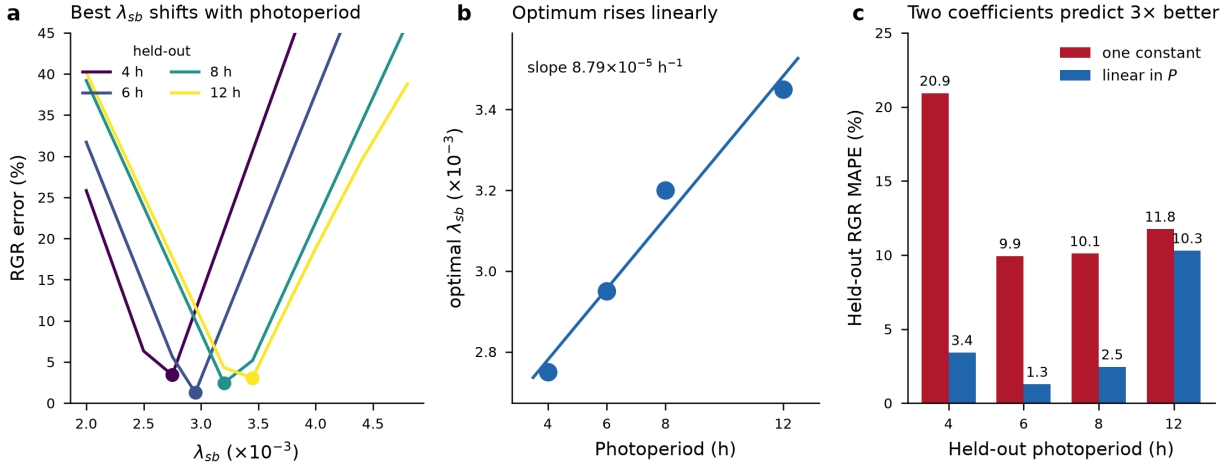


Figure 3: Held-out photoperiod prediction of relative growth rate. Regime A uses one shared growth coefficient; regime B lets the same coefficient vary linearly with photoperiod (Eq. 50). Each point is a photoperiod that was excluded from the fit that produced it.

6.3. Diel carbohydrate dynamics

The carbohydrate comparison uses one multiplicative scale per series, fitted by least squares before the error is computed, so the normalised RMSE scores the shape and the timing of the diel cycle. Over the reference series the mean starch nRMSE is 0.235 against the acceptance threshold of 0.25, with a bias of −3.1%. The mean sucrose nRMSE is 0.285 against a threshold of 0.50, with a bias of +10.7%. In the validation runs the largest per-photoperiod values are 0.219 for starch and 0.468 for sucrose. Figure 4 shows the traces.

The two series are not equally strong, and the fitted scales show why. For starch the scale is between 0.86 and 1.15 over the four fitted photoperiods, so shape and absolute level both agree and the starch nRMSE means what a reader would assume it means. For sucrose the scale is between 2.25 and 3.31: the simulated pool must be multiplied by a factor of two to three before it matches the measurements. Raw simulated sucrose maxima are 1.30 to 1.40 μmol C_6 gFW^{-1} against measured maxima of 3.18 to 4.71 μmol C_6 gFW^{-1}. The passing sucrose nRMSE of 0.285 therefore describes shape and timing only, and the absolute sucrose level is a known deficit of this version. We report the scale factor beside every carbohydrate error for that reason.

Within the fitted range the model reproduces the qualitative features of the diel cycle. Starch accumulates through the light period and degrades through the night along a near-linear trajectory. At dawn it returns to the level at which each run started, so the pool is close to empty at the end of every night. This behaviour follows from the drain law of Section 3.8, which targets the remaining pool over the remaining night, and it agrees with the near-linear nocturnal degradation reported for *Arabidopsis* [20, 45].

One quantitative feature is not reproduced. The measured end-of-day starch peak grows more steeply with photoperiod than the simulated peak. In the model the peak does rise with photoperiod, because the light period is longer, but the rise is weaker than measured. The model therefore under-represents how much a long day changes the size of the transient store, as opposed to its filling time. The same shortfall appears from the other side in Section 6.6, where assimilation per light hour is found to be constant.

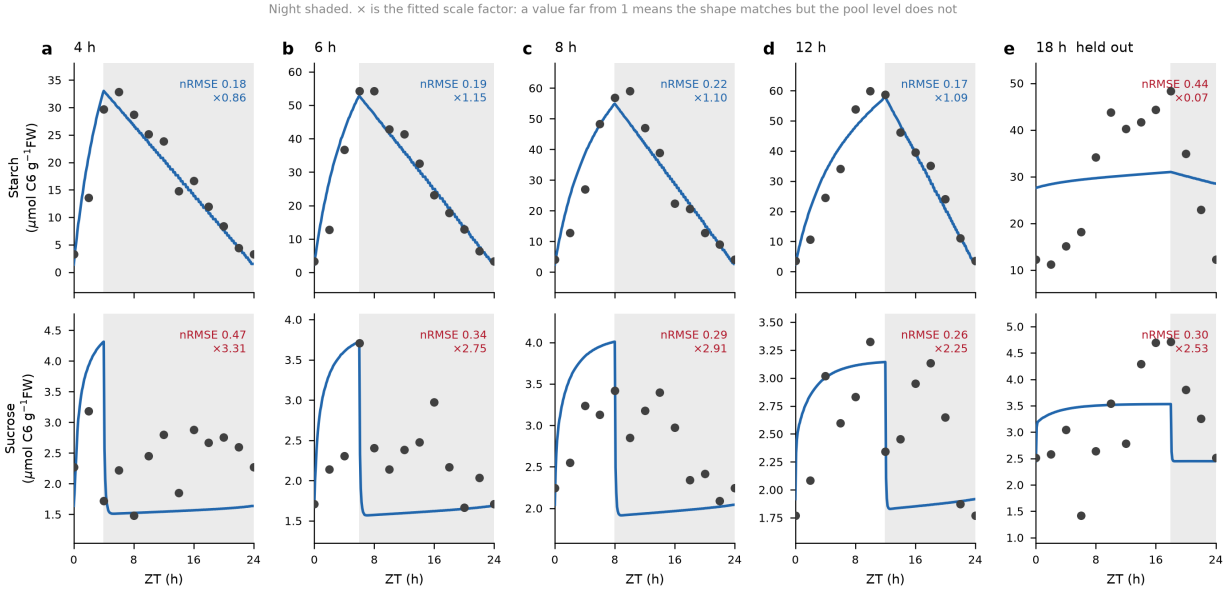


Figure 4: Diel starch and sucrose dynamics for the 4, 6, 8, 12 and 18 h photoperiods. Simulated traces are compared with the measured diel profiles of the repository reference file (see Section 5). Each panel prints the nRMSE and the fitted multiplicative scale. The scale is between 0.86 and 1.15 for starch at the four fitted photoperiods and between 2.25 and 3.31 for sucrose, so the sucrose comparison is one of shape and timing and not of absolute level. The 18 h column was not used in any fit; its starch nRMSE is not reported, because the fitted scale of 0.07 absorbs the divergence described in Section 6.6.

6.4. Ablation

The ablation compares three carbon-input pathways with the same downstream model, the same shared parameters and no refitting. In M0 the carbon input is prescribed. In M1 it is computed from Farquhar photosynthesis [15] at fixed stomatal conductance. In M2 the conductance follows the Jarvis formulation [25] and responds to the environment. Table 7 and Figure 5 give the result: the RGR error falls from 42.98% for M0 to 9.12% for M1 and to 8.58% for M2.

The large step is from prescribed to computed carbon input. M0 underestimates RGR at every photoperiod, by 41.2 to 45.0%, and its peak assimilation stays between 10.50 and 11.14 μmol gFW^{-1} h^{-1}. With computed assimilation the peak rises to 17.62 and the systematic underestimation disappears. The bias is not reduced but removed: M1 and M2 scatter on both sides of the targets, while M0 is low in all four scenarios. Parameters were frozen across the three variants, so the change comes from the carbon-input pathway and not from recalibration.

The Jarvis layer adds a smaller and unevenly distributed gain. Averaged over the four photoperiods it reduces the error from 9.12 to 8.58%, and the two variants differ most at the extremes of the range. At 4 h, M2 improves on M1 by 3.7 percentage points, from 14.86 to 11.21%. At 12 h it is worse by 2.0 points, 11.75 against 9.71%. At 6 h M2 is better by 2.9 points and at 8 h it is worse by 2.4 points. The mean gain is therefore small and its sign is photoperiod dependent. Adding environmental control of conductance lowers peak assimilation slightly, from 17.62 to 17.01 μmol gFW^{-1} h^{-1}, because the conductance multipliers are below one at the nominal condition. The value of M2 at a single nominal condition is therefore not a gain in RGR accuracy. Its value is that the carbon input responds to irradiance, humidity, temperature, ambient CO_2 and leaf water potential at all, which is what the sensitivity analysis of the next subsection exercises.

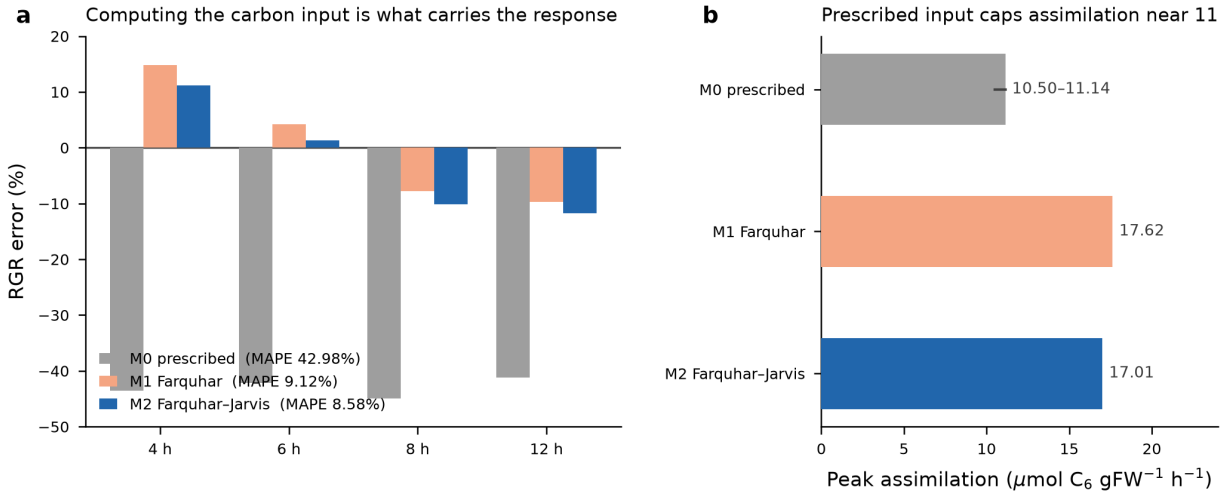


Figure 5: Contribution of the computed carbon input, with the shared parameters frozen in all three variants. (a) Signed RGR error per photoperiod. M0, which prescribes the carbon input, underestimates growth at every photoperiod by 41 to 45%; M1 and M2 scatter on both sides of the targets within ±15%, and their mean absolute errors of 9.12 and 8.58% are close enough that the Jarvis conductance layer is not what carries the improvement. (b) Peak assimilation reached by each variant, in the units of the carbon input. The prescribed form saturates near 11 μmol C_6 gFW^{-1} h^{-1} and varies by only 0.6 across the four photoperiods, while both computed forms reach 17; that ceiling is the mechanism behind the errors in panel (a).

6.5. Global sensitivity

The variance-based analysis follows Sobol' [52] with the estimator of Saltelli et al. [44], and the Morris screening follows Morris [39], Campolongo et al. [8]. Sixteen parameters were varied over the declared ranges of Appendix C. Table 8 gives the full ranking and Figure 6 shows the total-order indices next to the Morris measures.

Nutrient uptake capacity dominates the total-order ranking. The two largest total indices are 0.658 for maximum phosphorus uptake and 0.577 for maximum nitrogen uptake. Respiration frequency follows at 0.466, then ambient CO_2 concentration at 0.422 and leaf water potential at 0.412. The two Farquhar capacities come next, 0.352 for electron

transport rate and 0.334 for maximum carboxylation rate, then maximum PAR at 0.333. The ranking spans the nutrient, respiratory and gas-exchange blocks, as expected in a model where growth is limited by whichever supply is scarcest.

The main-effect share is small, and stating it correctly requires care. Summed over the sixteen parameters and averaged over the four photoperiods, the first-order indices give $\sum S_1 = 0.201$ against $\sum S_T = 4.675$. Because the first-order indices of a well-posed decomposition sum to at most one, the defensible statement is that parameters acting alone explain 20.1% of the variance in RGR, leaving 79.9% to interactions of two or more parameters. The ratio $1 - \sum S_1 / \sum S_T$ is not a variance share and we do not report one: $\sum S_T = 4.675$ exceeds unity precisely because every interaction is charged to each parameter involved in it, so the total-order indices double count by construction. What is well defined per parameter is the interaction remainder $(S_T - S_1)/S_T$, given in Table 8; at 12 h it is 1.000 for both nutrient uptake capacities, 0.998 for maximum PAR and 0.852 for leaf water potential. The Morris measures agree: 15 of the 16 parameters have $\sigma/\mu^* > 1$, and the exception is the one parameter with a total index at noise level. The two methods also rank the parameters consistently, with a Spearman rank correlation of $\rho = 0.944$ between Morris μ^* and Sobol' S_T. The model is interaction-dominated over the declared ranges: parameters act mainly through each other, not on their own.

A low main-effect share can also be an estimator artefact, so we checked convergence with sample size. As n rises from 512 to 4096, the number of first-order estimates clipped at zero falls from 8 to 3. Over the same range the summed first-order index stays between 0.151 and 0.340. The clipping decreases, which is what an estimator with shrinking variance should do, but the share does not increase with it. The low main-effect share is therefore a property of the model over these ranges and not an artefact of a small sample. All indices in Table 8 are reported unclipped, so small negative interaction remainders appear where an index is at noise level; we prefer that to hiding the estimator variance.

One entry deserves a comment. The growth coefficient λ_{sb}, the coefficient that Section 6.2 shows to carry the photoperiod response, has a total index of only 0.117 and ranks fifteenth of sixteen. There is no contradiction. A Sobol' index measures the variance contribution over a declared range at fixed forcing, while the cross-photoperiod result measures how a coefficient must change when the forcing changes. A parameter can be unimportant for variance at one condition and still be the parameter that fails to transfer between conditions.

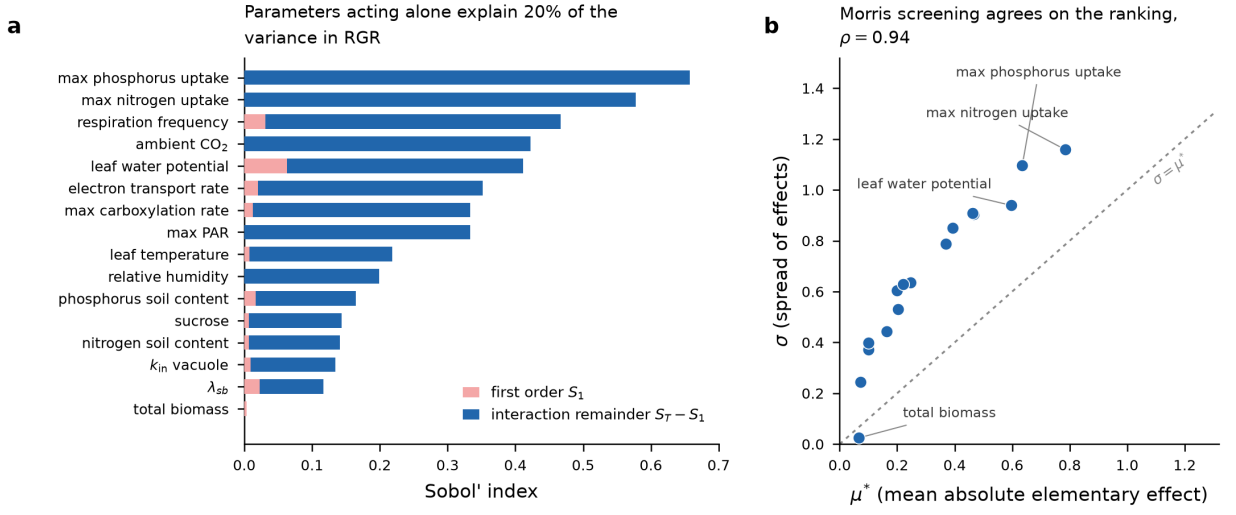


Figure 6: Global sensitivity of relative growth rate over the declared parameter ranges, averaged over the four photoperiods. (a) Total-order Sobol' index for each parameter, split into the first-order part and the interaction remainder $S_T - S_1$. Summed first-order indices are 0.201, so parameters acting alone account for about a fifth of the variance. The summed total-order indices come to 4.68: because a total-order index charges each interaction to every parameter involved in it, that sum exceeds unity and cannot be used as a denominator for a variance share. (b) Morris μ^* against σ for the same parameters. The two rankings agree at a Spearman correlation of 0.944, and 15 of the 16 parameters lie above the $\sigma = \mu^*$ line, which is the screening signature of effects that depend on where in the space they are evaluated.

6.6. Validity envelope

Two limits of the model were measured directly, and we report them as findings rather than as caveats. Figure 7 shows both.

The first limit is the 18 h photoperiod, which was never used in any fit. The sucrose comparison passes: nRMSE is 0.428 against the threshold of 0.50, with a fitted scale of 2.53, so the shape of the diel sucrose cycle survives three hours beyond the fitted range. The starch pool does not. Dawn starch is 401 μmol gFW^{-1} on day 30 and 1207 μmol gFW^{-1} on day 90, a net gain of about +14 per day that shows no sign of levelling off. Over the horizons tested the pool does not approach a level, it keeps gaining, and the distinction matters, because a wrong level can be rescaled while a divergence cannot. The fitted scale for 18 h starch is 0.07, which is the metric absorbing the divergence, so no 18 h starch nRMSE is quoted anywhere in this paper.

The mechanism is visible in the fluxes. The degradation cap is not binding at 18 h, so the failure is not a saturated drain. The drain law targets the remaining pool over the remaining night, so nightly export is proportional to what is left and to the reciprocal of the night length. Under a long day the pool filled during the light period grows while the night available to empty it shrinks. The day fill then outruns the drain, the pool carries a surplus into the next day, and the surplus compounds.

The second limit is the closure boundary, and it needs a criterion before it can be a number. We define the net daily starch gain as $\Delta a = a(\text{dawn}, d{+}1) - a(\text{dawn}, d)$ in μmol C_6 gFW^{-1}, evaluated at day $d = 30$ under the reference chamber forcing, and we call the cycle closed when $|\Delta a|$ stays below 0.5 μmol C_6 gFW^{-1} per day, which is 0.2% of the dusk pool at 12 h and an order of magnitude below the smallest gain that grows over the following sixty days. Under that criterion Δa is +0.300 at 12 h, +0.295 at 14 h and +0.389 at 15.0 h, rising to +0.547 at 15.5 h, +2.042 at 16 h, +13.780 at 18 h and +24.150 at 20 h. The cycle therefore closes to 15.0 h, is marginal at 15.5 h and fails from 16 h. That boundary sits three hours beyond the longest photoperiod used in any fit, which is a real margin and a narrow one.

The boundary is a property of the model and of the light dose together, not of the model alone. Under the shipped half-sine profile at 1200 μmol m^{-2} s^{-1} the same measurement gives +18.99 at 12 h, +32.56 at 14 h, +38.75 at 15 h, +45.00 at 16 h, +56.70 at 18 h and +67.68 at 20 h. Under that dose the balance never closes, not even at 12 h. Any statement of the form "the model closes the diel starch balance up to P hours" is therefore incomplete without the irradiance at which it was measured. We report the chamber column as the envelope of this model, because it is the condition with an experimental counterpart. The shipped column shows how strongly the boundary depends on the daily photon dose.

The third measurement is a bound rather than a boundary, and it explains the residual error of Section 6.1. Assimilation per hour of light is constant to six significant figures across the photoperiod series, so total daily assimilation is proportional to day length. The ratio of daily assimilation between the 12 and 4 h scenarios is therefore exactly 3.0000, while the reference RGR targets require a ratio of 3.82. Growth in this model is driven by the carbon input, so a supra-proportional factor of 1.27 has to come from somewhere else. Partitioning cannot supply it: the partition coefficients redistribute a fixed daily input between starch, sucrose and biomass, and they cannot make the total larger. This is the same gap that Eq. 50 closes empirically, seen at the level of the carbon budget instead of the fitted coefficient.

7. Discussion

7.1. How much each result is entitled to claim

The results carry different weight, and the difference is set by how much freedom the fit was given on each axis. We state the ranking first, because a reader should not have to reconstruct it from the sections. Four kinds of evidence appear in this paper, in order of increasing strength.

The weakest is the per-photoperiod growth-rate agreement. The saturated regime fits two coefficients at each photoperiod against two observations, so its near-zero error is a bisection residual and carries no predictive information (Section 6.1). We report it as a calibration diagnostic and nothing more. The shared regime is stronger but still in-sample: four coefficients are fitted once and reused at every photoperiod, so the 8.58% mean error measures consistency across conditions, not prediction. Next is the held-out cross-validation, where a whole photoperiod is excluded from the fit that predicts it (Section 6.2). This is a genuine prediction, and it shows both what the coupled dynamics generate on their own and what they miss. A photoperiod-independent growth coefficient predicts a held-out photoperiod with 13.18% error, and one linear coefficient reduces that to 4.36%. The missing photoperiod dependence is therefore real but compact, and localised to one growth-conversion coefficient.

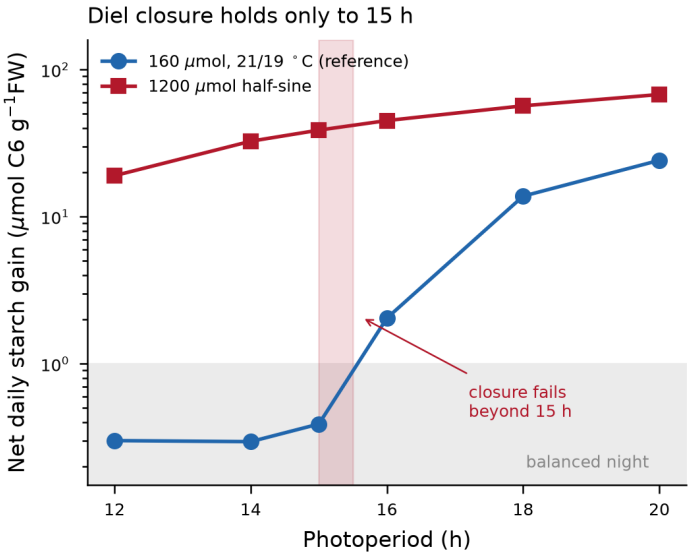


Figure 7: Net daily starch gain against photoperiod on a symmetric-log axis, under the chamber forcing and under the shipped half-sine profile at 1200 μmol m^{-2} s^{-1}. The band marks the closure boundary at 15.0–15.5 h under chamber forcing. Under the higher dose the balance does not close at any photoperiod shown, so the boundary depends on the light dose as well as on the model.

The strongest evidence is the axis on which nothing was fitted. No coefficient in the nutrient block was estimated against the nutrient response, so that block is entitled to judge the model. It fails in a specific way: simulated tissue nutrient concentration falls with increasing supply where the measurements rise (Section 7.6). The diel starch shape and the closure boundary sit alongside it, since the starch scale is near one and the 15.0–15.5 h boundary is a property of the drain law rather than of any fitted value. We draw the model's credibility from these unfitted axes, not from the growth-rate agreement. A model of this class earns trust where it can be wrong and is not, and we have tried to make those places easy to find rather than easy to miss.

7.2. What the coupled formulation buys

The ablation puts a number on the coupling: computing the carbon input rather than prescribing it removes a bias present at all four photoperiods, with no parameter refitted (Section 6.4). The reason is that a prescribed maximum rate must be chosen once, and a single choice cannot be correct at four day lengths, because the plant it describes has a different leaf area and sucrose status in each case. The computed input is a function of the states and the forcing, so it changes with them.

The Jarvis layer should be read differently. It reduces the mean error from 9.12 to 8.58%, which is a small change, and it moves the error in opposite directions at the two ends of the photoperiod range. Reporting that as an accuracy gain would misstate it. What the layer provides is a carbon input that responds to irradiance, humidity, temperature, ambient CO_2 and leaf water potential. The sensitivity analysis shows this is not a formality: ambient CO_2 has a total-order index of 0.422, leaf water potential 0.412 and maximum PAR 0.333. In M1 those three parameters would move nothing through the conductance path. A model intended for use across environments needs that path even when it does not improve a single nominal-condition metric, and a model evaluated only at its calibration point cannot show the difference.

This is where the model sits relative to existing families. It is not a field-scale crop simulator, and it does not resolve architecture as a functional-structural model does. It is a compact whole-plant core in which the carbon input is computed from the environment and then propagated explicitly into carbohydrate, nutrient and biomass states, with the numerical treatment and the evaluation protocol stated. Many whole-plant and crop simulators prescribe that input [22, 26, 64], and the ablation measures what prescribing it costs in this setting. The physiological submodels are inherited [15, 25, 6]; the coupling, the analytic closure and the evaluation are what this paper contributes.

7.3. Parameter transferability and what the design exposed

The cross-photoperiod result is the part of this work with the widest reach beyond plant modelling. One shared growth coefficient predicts a held-out photoperiod with 13.18% error; the same coefficient made linear in photoperiod predicts it with 4.36%. Both regimes use the same equations, the same states and the same solver. The difference is entirely in how one coefficient is allowed to depend on a condition of the experiment.

That is a statement about parameter transferability, and it is a common failure mode in simulation practice. A coefficient fitted at one operating condition is usually reported as a property of the model, and it is then applied at other conditions as if it were one. When it is not, the model appears to be structurally wrong, because the error shows up in the state trajectories rather than in the parameter. Our results separate the two cases. The gas-exchange layer, the carbohydrate subsystem and the nutrient subsystem all transfer with fixed coefficients across 4 to 12 h. One coefficient in the growth-conversion term does not, and it needs one extra degree of freedom to do so.

The design is what made this visible. A random train-test split over all available observations would have placed neighbouring photoperiods on both sides of the split, and interpolation between them would have hidden the dependence. Holding out an entire condition asks the question that matters for a simulator: does the fitted model work at a condition it has never seen? We suggest leave-one-condition-out evaluation as a default for models calibrated across a treatment series. The reason is not that it is more severe. It is the design that localises a transferability failure to a specific coefficient. A single held-out error tells you the model missed. A sweep of held-out folds, each with its own refitted coefficient, tells you which coefficient moved and in which direction, and here it moved monotonically with photoperiod.

Two properties of the design make it reusable outside this model. First, the held-out condition must be a condition and not a random subset, because random splitting lets neighbouring treatments substitute for each other. Second, one coefficient at a time should be released. Regime B differs from regime A by a single degree of freedom, so the improvement from 13.18 to 4.36% can be attributed to that coefficient and to nothing else. Releasing several coefficients at once would have produced a similar error reduction and located nothing. Together the two properties turn a fit quality measurement into a diagnostic that names a term in the model, which is what a modeller needs before deciding whether to recalibrate or to reformulate.

The physiological reading must stay careful. A growth coefficient that rises with day length means that the conversion from loaded sucrose to structural biomass becomes more efficient as the photoperiod lengthens. Increased sink activity under long days is a plausible candidate [51, 54], and so is a change in the ratio of fresh weight to carbon, and so is a shift in allocation between leaf and root. The present data cannot separate these, and one linear coefficient fitted over four photoperiods will fit all three explanations equally well. We therefore report the dependence as a measured property of the calibration and not as an identified mechanism.

7.4. What the carbohydrate results do and do not show

The two carbohydrate series should be read with different confidence. For starch the fitted scales lie between 0.86 and 1.15, so the comparison covers level as well as shape. A mean nRMSE of 0.235 against a threshold of 0.25 can therefore be taken at face value. For sucrose the scales lie between 2.25 and 3.31, so the same style of statement would be misleading. The passing value of 0.285 means that the model puts the sucrose maxima and minima at the right times, with roughly the right relative depth. It says nothing about the absolute pool size, which is a factor of two to three too low.

The distinction matters beyond this model, because a scale-fitted error metric is common when reference data are indirect or on a different basis. Fitting one multiplicative scale is defensible: it separates a units or basis mismatch from a dynamic error, and it is the only way to compare a normalised model pool against a measured tissue concentration. But the fitted scale is part of the result and must be reported with it. A scale near one and a scale near three support different claims from the same nRMSE, and only the first supports a statement about level.

7.5. Interaction dominance and its consequences for calibration

Over the declared parameter ranges, parameters acting alone explain about 20% of the variance in RGR, so roughly four fifths of it arises from interactions. In addition, 15 of the 16 parameters have a Morris σ/μ^* above one. Two independent methods agree on the ranking at $\rho = 0.944$, and the convergence check shows the low main-effect share is stable as the sample grows. The conclusion is not that main effects are absent, but that they are small relative to interactions across the tested domain.

This has a practical consequence for anyone calibrating a model of this class. One-at-a-time tuning is unreliable here. Suppose a parameter's own effect explains a small part of the variance and most of its influence runs through interactions. The sensitivity measured by moving that parameter alone then depends on where the other fifteen happen to sit. Tuning it at one operating point can make behaviour worse at another, and the sequence in which parameters are tuned changes the result. The structure of the model explains why. Growth is limited by whichever supply is scarcest, and the largest total-order indices belong to the nutrient uptake capacities, 0.658 and 0.577. A nutrient parameter changes which limitation binds, and that changes how strongly every gas-exchange parameter matters. The interaction is the model working as designed, and it means calibration should be done jointly, with a global method, over a declared domain.

Two limits on this reading should be kept. The indices are conditional on the ranges in Appendix C; wider ranges would raise the interaction share and narrower ones would lower it. And they describe model behaviour, not plant physiology. That nutrient uptake capacity dominates variance in this model is a statement about this model over these ranges.

7.6. Limitations

Six limitations bound what can be concluded, and each is measured rather than asserted.

Four coefficients were fitted. The growth coefficient, the two starch partitioning rates and the half-activation constant of the sink term were estimated against the reference data, so the in-sample results of Section 6.1 are not independent tests. This is a large improvement on the eight-per-photoperiod calibration of the earlier version, which had zero degrees of freedom against eight observations, but four fitted coefficients are still four. The held-out results of Section 6.2 and the 18 h evaluation of Section 6.6 are the parts of the evaluation that no fitting touched.

The carbohydrate metric is scale-fitted, and for sucrose the fitted scale is large. The starch scales lie between 0.86 and 1.15, so the starch comparison covers level as well as shape. The sucrose scales lie between 2.25 and 3.31, so the simulated pool is a factor of two to three below the measurements in absolute terms and the passing nRMSE of 0.285 evaluates shape and timing only. Raw simulated maxima are 1.30 to 1.40 μmol C_6 gFW^{-1} against measured maxima of 3.18 to 4.71 μmol C_6 gFW^{-1}. The deficit is one of the clearest targets for the next version. It should also be read together with the whole-tissue basis of the measurements. The model quantity compared here is the sum of the cytosolic and vacuolar pools, so a partition error between the two would not show in the total.

Some parameters remain unsourced. The provenance ledger classifies 14 of 124 entries as having no traceable origin and 15 as fitted, of which four were calibrated in this work, and both groups are listed in Appendix B with that status attached. Unsourced constants are a work list and not a defence; we report them so that a reader can see which numbers rest on a measurement and which do not.

The diel starch balance closes only to 15.0–15.5 h under chamber forcing, and under the higher shipped light dose it does not close at any photoperiod tested. The model is therefore not valid for long-day simulation as it stands, and any application beyond the closure boundary would need the drain law changed first.

The RGR response to photoperiod is bounded by construction. Assimilation per light hour is constant to six significant figures, so daily assimilation is proportional to day length, and the 12 to 4 h ratio is exactly 3.0000 where the targets need 3.82. No choice of partitioning parameters can produce the missing factor of 1.27, because partitioning redistributes a fixed daily input. The linear growth coefficient covers this gap empirically, and the underlying bound remains.

Simulated tissue nutrient concentration falls with increasing supply where the measurements rise. The nutrient block is out of sample, no coefficient in it was fitted to the nutrient response, so it is entitled to judge the model and the discrepancy is reported and not tuned away. The likely cause is in the structure of the nutrient balance rather than in a parameter value. Every term in the internal nutrient equation is bounded, and the concentration is consumed in proportion to assimilation. A supply increase raises assimilation, which raises consumption, and with a bounded uptake term the pool can fall. Resolving this requires the uptake and consumption terms to be reformulated, which is beyond a recalibration.

7.7. Pushing the envelope past 15.5 hours

The closure failure has a specific cause, so the candidate remedies are also specific. The nocturnal drain targets the remaining pool over the remaining night. Its rate therefore scales with the reciprocal of night length, while the daytime fill scales with day length, and beyond about 15 h the fill wins. Three changes could plausibly move the boundary, and each is testable within the present model.

The first is to make the daytime partition to starch respond to the size of the starch pool relative to what the coming night can export. In the current formulation the partition coefficient responds to sucrose status, not to the starch balance, so nothing tells the plant to stop filling a store it cannot empty. Measurements support such a dependence, since *Arabidopsis* adjusts partitioning to starch across photoperiods [54, 36]. The second is to raise the degradation capacity. The cap is not binding at 18 h, so it cannot be the limit on its own. A higher cap combined with a drain law that also depends on the pool level would extend the range over which the drain can match the fill. The third is to allow the peak amplitude of the store to grow with photoperiod, which is also the feature that Section 6.3 identifies as under-represented against the measurements. That the same shortfall shows up in the diel comparison and at the closure boundary suggests one structural cause rather than two.

We do not claim any of these will work. Each adds a coefficient that would have to be identified against data, and the measurements available at 18 h are limited. A change that closes the balance could easily be a change that fits one series. What we can state is narrower. The boundary sits at 15.0–15.5 h under chamber forcing, and its cause lies in the drain law and its interaction with the day fill. Testing the three candidates above is the next step, not a result.

8. Conclusion

We presented a whole-plant simulation model in which environmental forcing is converted into a photosynthetic carbon input through Jarvis stomatal conductance and Farquhar C3 photosynthesis. That carbon input is then propagated through diel carbohydrate, nutrient and biomass dynamics. The supply and demand sides of gas exchange are closed analytically at runtime as a quadratic in the intercellular CO_2 concentration, so the operating point is obtained without an inner iteration. The eleven slow states are advanced by an adaptive integrator. The formulation, its numerical treatment and the evaluation protocol are the contribution; the physiological submodels are inherited.

The evaluation used four shared fitted coefficients rather than the eight-per-photoperiod calibration of the earlier version. In sample the model reproduces the relative growth rate across four photoperiods with a mean absolute percentage error of 8.58%. In a leave-one-photoperiod-out test, one shared growth coefficient predicts a held-out photoperiod less well than the same coefficient made linear in photoperiod, which reaches 4.36% with two coefficients. The ablation shows what the coupling contributes: replacing a prescribed carbon input with a computed one moves the error from 42.98 to 8.58% with parameters frozen. The variance-based analysis shows an interaction-dominated model over the declared ranges, led by nutrient uptake capacity, respiration and gas-exchange capacity. Calibration of this class of model should therefore be done jointly, not one parameter at a time.

The paper also states where the model stops working. Under the chamber forcing the diel starch balance closes up to a photoperiod of 15.0 to 15.5 h and fails beyond it. Under a higher light dose it does not close at all, so the boundary belongs to the model and the light dose together. At 18 h the sucrose cycle still has the right shape while the starch pool continues to accumulate over the whole test horizon. Assimilation per light hour is constant, which bounds the growth response to photoperiod below what the reference data require, and the linear growth coefficient covers that gap empirically rather than mechanistically. The absolute sucrose level is a factor of two to three too low, so the sucrose comparison is one of shape and timing.

Three directions follow. The drain law and the daytime partition to starch are the place to look for a formulation that closes the balance at long days. The absolute sucrose level and the nutrient concentration response are two independent out-of-sample discrepancies that point at the structure of the sucrose and nutrient balances rather than at their coefficients. And the leave-one-condition-out design used here can be applied to other conditions, irradiance and temperature first, to test whether the remaining coefficients transfer as well across those as they do across photoperiod.

A. Boundary cases of the gas-exchange closure

Section 3.6 solves the supply–demand closure as a quadratic in C_i and returns the admissible positive root on each branch. Two limiting conductance states need a separate treatment. Both have physical answers rather than clamps, and neither is exercised in any reported run, but stating them completes the definition of the closure.

The vanishing-conductance case has a physical answer rather than a clamp, and the one-sided limit and the closed boundary are two different states. As $g_c \to 0^+$ the quadratic degenerates: $a \to 0$ and $b \to R_d d_1 - \text{num}$, and the finite root of the remaining linear equation is

$$\lim_{g_c \to 0^+} C_i = \frac{\mathrm{num}\,\Gamma^* + R_d d_0}{\mathrm{num} - R_d d_1}, \tag{55}$$

provided the denominator is positive, which holds for both branches at the parameter values used here. Substituting this root into Eq. (18) gives $A_b = R_d$ and hence $A_n \to 0$: supply and demand meet at the compensation point, where diffusion delivers nothing and gross assimilation exactly pays respiration. The limit is a finite C_i, not Γ^*. At the production values it evaluates to 67.94 μmol mol^{-1} on the Rubisco-limited branch and 53.35 on the electron-transport-limited branch, and a numerical sweep in g_c reproduces both.

Exactly closed stomata are a separate case. At $g_c = 0$ the supply relation carries no information about C_i, the quadratic is empty, and the physical state is a leaf that admits no CO_2 and only respires. The implementation returns that state directly at the boundary $g_c \le 0$,

$$C_i = \Gamma^*, \qquad A_n = -R_d, \tag{56}$$

which is the dark limit and replaces a clamp on the conductance.

This assignment is an implementation boundary convention and not the continuous extension of Eq. (55). The one-sided limit derived above gives $A_n \to 0$ as $g_c \to 0^+$, whereas the convention returns $A_n = -R_d$ at $g_c = 0$, so the two differ by R_d at the boundary. We state the difference rather than smooth it over. The convention is the physically meaningful one for a leaf with no gas exchange, since gross assimilation requires a CO_2 supply that a shut stoma does not provide, while the limit describes a leaf whose stomata are almost shut and which is still exchanging enough CO_2 to break even. Whether the discrepancy matters is a question about the trajectory: the conductance model of Section 3.4 has a positive floor on its light response, so $g_c = 0$ is never reached in any reported run and the branch is not exercised. Respiration in darkness is carried by the same expression that carries assimilation in the light.

One numerical caveat belongs with this derivation. Written in the form above the quadratic loses significance when $a = -g_c d_1$ becomes comparable to the rounding error of b, which at these parameter values occurs below $g_c \approx 10^{-8}$; a sweep to $g_c = 10^{-11}$ returns a spurious positive A_n. The conductance model of Section 3.4 has a positive floor on its light response, so no reported run enters that range, and we record the bound rather than rely on it being unreachable.

B. Parameter provenance ledger

Every numerical constant in the released model carries a provenance record in `config/parameter_provenance.csv`, reproduced in full as Table S1 of the Supplementary Material. The tier states how the value was obtained: T1 measured, T2 derived, T3 inherited, T4 constrained, T5 fitted, T6 unsourced. A T6 entry is a value that we could not trace to a measurement, a derivation or the source model, and it is declared as such rather than presented as sourced. The ledger holds 124 entries, of which 4 are measured (T1), 55 are derived (T2), 27 are inherited (T3), 9 are constrained (T4), 15 are fitted (T5) and 14 are unsourced (T6). Every key of the released configuration appears in the ledger, as does every constant hard-coded in the model source, and every ledger value that names a configuration key matches the value the reported runs load; these properties are checked by script rather than by reading. Two rows are attributed to this work rather than to a publication, because no external source exists for them: the growth coefficient obtained by the held-out sweep and the assimilation scale constrained by the reference data.

The complete ledger of all 124 entries, with value, unit, tier, source and locator for each, is provided as Table S1 in the Supplementary Material and is released with the code as `config/parameter_provenance.csv`. The curated physical parameter table in the main text (Table 2) is the subset most relevant to reading the model equations.

C. Sensitivity analysis: parameters, ranges and sampling scales

Table 9 is generated from `config/sensitivity_params.csv` and gives the factors varied in the sensitivity study together with their ranges, sampling scales and the role each factor plays in the model. All reported Sobol' and Morris indices are conditional on exactly these ranges: an index is a statement about output variance over a declared input distribution, and it is not transferable to a different one. A factor sampled on a logarithmic scale is varied uniformly in $\log_{10}$, which is the appropriate choice for a rate constant spanning more than one order of magnitude. Of the 20 factors listed, 16 are active in the reported design and 4 are disabled; the reasons for disabling are given in the table.

Declaration of competing interest

The authors declare no competing interests.

Data and code availability

The simulation code, configuration files, reference data and the scripts that generate every table and figure in this paper are available at [REPOSITORY URL AND DOI]. The parameter provenance ledger reproduced as Table S1 in the Supplementary Material is released with the code as `config/parameter_provenance.csv`, and the numerical verification, convergence and carbon-closure studies are released as the documents cited in Section 4. The diel carbohydrate reference data are redistributed from the supplementary material of Sulpice et al. [54] under the terms of that publication.

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Table 2
Model parameters, generated from the provenance ledger of the released repository. Values are those loaded by the shipped configuration, or the literal in the source where a parameter is not exposed in the configuration file. Tiers: T1 measured, T2 derived, T3 inherited from the source model, T4 constrained, T5 fitted, T6 unsourced. The four parameters calibrated in this work are given separately in Table 3. [a]Literal in the source, not exposed as a configuration key; its ledger row records the same provenance fields as a configuration parameter.

Symbol	Meaning	Value	Unit	Tier	Source
f	Photoperiod length	8	h	T1	Sulpice et al. [54]
$\Phi_{c,\max}$	Reference assimilation rate	12.7	μmol C_6 gFW^{-1} h^{-1}	T3	Tedone [55]
$B_{l,\min}$	Leaf-biomass viability threshold	0.001	gFW	T4	–
λ_c	Residual assimilation at full starch	1	–	T5	–
C_a	Ambient CO_2 concentration	400	μmol mol^{-1}	T4	–
g_s^{fix}	Fixed conductance (M1 configuration)	0.1	mol m^{-2} s^{-1}	T5	–
SLA	Specific leaf area	0.0055	m^2 gFW^{-1}	T1	Tolleter et al. [59]
Γ^*	CO_2 compensation point	42	μmol mol^{-1}	T6	–
$J_{\max}$	Maximal electron transport rate	90	μmol m^{-2} s^{-1}	T5	–
$V_{c\max}$	Maximal carboxylation rate	45	μmol m^{-2} s^{-1}	T5	–
K_c	Rubisco Michaelis constant, CO_2	404.9	μmol mol^{-1}	T3	Bernacchi et al. [4]
K_o	Rubisco Michaelis constant, O_2	278.4	mmol mol^{-1}	T3	Bernacchi et al. [4]
O	Oxygen concentration	210	mmol mol^{-1}	T4	–
T_{ref}	Kinetic reference temperature	25	°C	T2	normalisation[a]
$T_{\text{day}}, T_{\text{night}}$	Chamber leaf temperature	21, 19	°C	T1	Sulpice et al. [54]
RH	Relative humidity	60	%	T6	–
ψ_w	Leaf water potential	-0.6	MPa	T5	–
R_d	Dark respiration rate	1.5	μmol m^{-2} s^{-1}	T2	–
ϕ	Apparent quantum yield	0.385	mol e$^-$ mol^{-1}	T6	–
θ	Light-response curvature	0.7	–	T3	Farquhar and Wong [16]
b_1	Jarvis light-response scale	0.6	–	T6	–
b_2	Jarvis light-response slope	0.002	–	T6	–
$D_{a,\max}$	Maximal starch degradation rate	6	μmol C_6 gFW^{-1} h^{-1}	T3	Tedone [55]
$a_{\min}$	Non-mobilised starch residue	0.15	μmol C_6 gFW^{-1}	T3	Tedone [55]
$s_{\max}$	Sucrose half-saturation, excess	4.712	μmol C_6 gFW^{-1}	T3	Tedone [55]
$s_{\min}$	Sucrose half-saturation, deficit	1.3	μmol C_6 gFW^{-1}	T3	Tedone [55]
λ_{sni}	Night partition increase rate	0.12	h^{-1}	T5	–
λ_g	Loading efficiency floor	0.65	–	T5	–
r_m	Maintenance respiration rate	0.79	h^{-1}	T3	Tedone [55]
η	Phloem loading frequency	1.98	h^{-1}	T3	Tedone [55]
τ_c	Transport cost fraction	0.035	–	T4	Tedone [55]
k_{in}	Vacuolar influx rate	0.1	h^{-1}	T5	ratio from Seydel et al. [47]
k_{out}	Vacuolar efflux rate	0.041959	h^{-1}	T2	Seydel et al. [47]
n_H	Sink-activation Hill exponent	3	–	T4	this work
Y	Construction cost of biomass	0	μmol C_6 gFW^{-1}	T6	disabled, see text
μ_l	Leaf mortality rate	5×10^{-6}	h^{-1}	T3	Tedone [55]
μ_r	Root mortality rate	5×10^{-6}	h^{-1}	T3	Tedone [55]
γ_l	Leaf self-competition coefficient	5×10^{-6}	gFW^{-1} h^{-1}	T3	Tedone [55]
γ_r	Root self-competition coefficient	5×10^{-6}	gFW^{-1} h^{-1}	T3	Tedone [55]
ω	Optimal N:P ratio	10	–	T3	Tedone [55]
$r_{A,N}$	N cost of carbon turnover	0.0460308	μmol N (μmol C_6)$^{-1}$	T2	Tedone [55]
$r_{A,P}$	P cost of carbon turnover	0.00460308	μmol P (μmol C_6)$^{-1}$	T2	Tedone [55]
N_{ph}	N threshold for assimilation	5.75	μmol N gFW^{-1}	T3	Tedone [55]
P_{ph}	P threshold for assimilation	0.575	μmol P gFW^{-1}	T2	Tedone [55]
ν	Nutrient conversion parameter	0.066	–	T5	–
N_s	Soil nitrogen concentration	15	μmol cm^{-3}	T4	Tedone [55]
P_s	Soil phosphorus concentration	0.15	μmol cm^{-3}	T3	Tedone [55]
$I_{N,\max}$	Maximal N uptake rate	6.44	μmol N gFW^{-1} h^{-1}	T3	Tedone [55]
$I_{P,\max}$	Maximal P uptake rate	0.4	μmol P gFW^{-1} h^{-1}	T3	Tedone [55]
K_N	Soil N half-saturation	0.125	μmol cm^{-3}	T3	Tedone [55]
K_P	Soil P half-saturation	0.006736	μmol cm^{-3}	T3	Tedone [55]
λ_k	Stoichiometric imbalance gain	50	–	T5	–
c_N	Sucrose cost of N uptake	0.68	μmol C_6 (μmol N)$^{-1}$	T3	Tedone [55]
c_P	Sucrose cost of P uptake	0.308	μmol C_6 (μmol P)$^{-1}$	T3	Tedone [55]
N_s^{ref}	Soil N reference (hard-coded)	12	μmol cm^{-3}	T6	–
P_s^{ref}	Soil P reference (hard-coded)	0.15	μmol cm^{-3}	T6	–
$g_{s,\max}$	Maximal stomatal conductance	$1/\sqrt{2}$	mol m^{-2} s^{-1}	T2	this work[a]
b_{10}	Jarvis residual light conductance	0.05	–	T6	–[a]
b_5	Jarvis VPD slope	0.2	kPa^{-1}	T6	–[a]
b_6	Jarvis water-potential slope	3.0	MPa^{-1}	T6	–[a]
b_8	Jarvis CO_2 response floor	0.2	–	T6	–[a]
T_0	Jarvis temperature optimum	25	°C	T3	cardinal set[a]
T_1, T_h	Jarvis temperature zeros	5, 45	°C	T3	cardinal set[a]
$\psi_{\min}$	Jarvis stomatal closure potential	−1.5	MPa	T3	wilting bound[a]

Table 3
The four parameters calibrated in this work. All four are shared: one value of each is used at every photoperiod in every reported run. All are tier T5 by the classification of Section 3.12. The legacy baseline calibrated eight parameters per photoperiod against the same eight observations. The ledger of Appendix B marks 15 entries as tier T5. That tier records that a value was obtained by fitting at some point in the model's history, including values inherited already fitted from the source model; the four listed here are the ones calibrated in this work, and the remaining T5 entries were not adjusted for any result reported in this paper.

Symbol	Meaning	Value	Unit	Calibrated against
λ_{sb}	Biomass conversion coefficient	0.00295	gFW (μmol C_6)$^{-1}$	Relative growth rate as a function of photoperiod, over the four measured photoperiods jointly (Fig. 2)
λ_{sdr}	Day partition reduction rate	0.50	h^{-1}	Shape and timing of the diel starch trace, jointly with λ_{sdi} (Fig. 4)
λ_{sdi}	Day partition increase rate	0.12	h^{-1}	Shape and timing of the diel starch trace, jointly with λ_{sdr} (Fig. 4)
K_s	Sink-activation half-saturation	0.90	μmol C_6 gFW^{-1}	Invariance of the sucrose level across the photoperiod series (Fig. 4)

Table 4
Parameterisation regimes. All three use the same chamber forcing (PAR 160 μmol m^{-2} s^{-1}, square wave, 21/19 °C), the same initial state and the same sourced parameters. P denotes photoperiod in hours. The observation count is two per photoperiod, the published RGR and the published end-of-day starch. Regime R1 is reported as a calibration diagnostic; only R2 and R3 enter the cross-validation of Section 5.4.

Regime	**Fitted coefficients**	**Fitted** values	**Obs.** used	**Role**
R1 saturated	λ_{sb}, λ_{sdi}, per photoperiod	8	8	calibration diagnostic
R2 shared	λ_{sb}, λ_{sdr}, λ_{sdi}, sink constant	4	8	held-out prediction
R3 shared, linear	as R2 with $\lambda_{sb} = c_0 + c_1 P$	5	8	held-out prediction

Table 5
Relative growth rate under the shared parameterisation, chamber forcing, 30 days. Targets are the *Arabidopsis thaliana* rosette values of Sulpice et al. [54]. The last column gives the per-photoperiod residual of the saturated per-photoperiod calibration on the grid described in Section 6.1; those residuals are set by the grid resolution and are reported as calibration diagnostics, not as errors of prediction.

Photoperiod (h)	Target RGR (d^{-1})	Simulated RGR (d^{-1})	Error (%)	Saturated fit residual (%)
4	0.0680	0.0756	+11.21	3.42
6	0.1135	0.1149	+1.28	1.28
8	0.1708	0.1536	−10.10	2.45
12	0.2600	0.2294	−11.75	3.03
Mean absolute percentage error			8.58	–

Table 6
Leave-one-photoperiod-out prediction of relative growth rate. In regime A the growth coefficient is a single shared constant; in regime B it is linear in photoperiod with two coefficients (Eq. 50). Train error is the mean over the three fitted photoperiods; test error is the held-out photoperiod. The λ_{sb} column gives the value applied at the held-out photoperiod.

Regime	Held out (h)	λ_{sb}	Train MAPE (%)	Test MAPE (%)
A: shared constant	4	0.00320	5.57	20.93
A: shared constant	6	0.00320	9.24	9.94
A: shared constant	8	0.00295	8.08	10.10
A: shared constant	12	0.00295	7.53	11.75
Regime A, mean held-out error				13.18
B: linear in P	4	0.00275	2.25	3.42
B: linear in P	6	0.00295	2.97	1.28
B: linear in P	8	0.00320	2.57	2.45
B: linear in P	12	0.00370	2.38	10.30
Regime B, mean held-out error				4.36

Table 7
Ablation of the carbon-input pathway. M0 prescribes the carbon input, M1 computes it from Farquhar photosynthesis at fixed conductance, and M2 adds Jarvis stomatal conductance. Shared parameters are frozen across the three variants. Errors are relative errors in RGR; the last column is the peak assimilation rate.

	RGR error (%) by photoperiod				MAPE	Peak Φ_c
Variant	4 h	6 h	8 h	12 h	(%)	(μmol gFW^{-1}h^{-1})
M0 prescribed	43.53	42.18	44.97	41.22	42.98	10.50–11.14
M1 Farquhar	14.86	4.18	7.73	9.71	9.12	17.62
M2 Farquhar–Jarvis	11.21	1.28	10.10	11.75	8.58	17.01

Table 8
Global sensitivity of relative growth rate, averaged over the four photoperiods. S_1 is the first-order and S_T the total-order Sobol' index; the interaction remainder is $S_T - S_1$. μ^* and σ are the Morris measures. Indices are reported unclipped, so a remainder at noise level can be slightly negative. Ranges and sampling scales are in Appendix C; the indices are conditional on those ranges.

Parameter	S_1	S_T	$S_T - S_1$	μ^*	σ/μ^*
max_phosphorus_uptake	0.000	0.658	0.658	0.635	1.73
max_nitrogen_uptake	0.000	0.577	0.577	0.785	1.48
respiration_frequency	0.031	0.466	0.435	0.466	1.94
ambient_CO2_concentration	0.000	0.422	0.422	0.370	2.13
leaf_water_potential	0.063	0.412	0.349	0.597	1.57
electron_transport_rate	0.020	0.352	0.332	0.247	2.57
max_rubisco_carboxylation_rate	0.013	0.334	0.321	0.393	2.16
max_PAR	0.000	0.333	0.333	0.463	1.96
leaf_temperature	0.008	0.219	0.211	0.198	3.05
relative_humidity	0.000	0.199	0.199	0.222	2.84
phosphorus_soil_content	0.017	0.164	0.148	0.202	2.62
sucrose (initial)	0.007	0.144	0.137	0.164	2.70
nitrogen_soil_content	0.007	0.141	0.134	0.100	3.72
k_vacuole_in	0.009	0.135	0.126	0.101	3.96
lambda_sb	0.023	0.117	0.094	0.073	3.32
total_biomass (initial)	0.003	0.001	−0.002	0.066	0.37
Sum	0.201	4.675			

Table 9
Sensitivity analysis design, generated from `config/sensitivity_params.csv`. Role distinguishes a model parameter from an environmental driver, an initial condition and a physical constant. The basis column records why the range was chosen.

Factor	Lower	Upper	Scale	Role	Basis
Active in the reported design					
`growth.lambda_sb`	0.002	0.01	linear	model parameter	Calibrated; ±70% around default 0.00587 (Schulz et al. 2019)
`carbohydrates.respiration_frequency`	0.4	1.2	linear	model parameter	Calibrated; ±50% around default 0.79
`nutrients.max_nitrogen_uptake`	1	15	log	model parameter	Literature range for herbaceous crops (Barber 1995; Lambers et al. 2008)
`nutrients.max_phosphorus_uptake`	0.05	1.5	log	model parameter	Literature range for herbaceous crops (Barber 1995)
`nutrients.nitrogen_soil_content`	3	30	linear	environment	Low-nutrient to high-nutrient agricultural soil (Brady & Weil 2010)
`nutrients.phosphorus_soil_content`	0.03	0.5	log	environment	Low to high-P agricultural soil (Brady & Weil 2010)
`gas_exchange.leaf_temperature`	10	35	linear	environment	Controlled environment range; cold stress to heat stress
`gas_exchange.relative_humidity`	30	90	linear	environment	Arid to humid growing conditions
`gas_exchange.ambient_CO2_concentration`	280	800	linear	environment	Pre-industrial to RCP 8.5 projected 2100 (Meinshausen et al. 2011)
`gas_exchange.leaf_water_potential`	-1.5	-0.1	linear	environment	Severe water stress to turgid (Hsiao 1973; Jones 2014)
`gas_exchange.max_rubisco_carboxylation_rate`	20	120	log	model parameter	C3 herbaceous range (Wullschleger 1993; Medlyn et al. 2002)
`gas_exchange.electron_transport_rate`	50	200	log	model parameter	C3 herbaceous range (Wullschleger 1993)
`photo.max_PAR`	300	2000	linear	environment	Shade canopy (300) to full sun (2000 μmol mš sź) (McCree 1972)
`carbohydrates.sucrose`	0.8	3	linear	initial condition	Observed range in spinach seedlings (% DW; Geiger & Servaites 1994)
`growth.total_biomass`	0.01	0.12	linear	initial condition	Seedling (0.01 g FW) to established juvenile plant (0.12 g FW)
`carbohydrates.k_vacuole_in`	0.005	0.4	linear	model parameter	Cytosolvacuole exchange rate; covers calibrated per-PP range 0.011-0.230 h-1 (Lohaus 2007)
Disabled					
`carbohydrates.lambda_sni`	0.04	0.18	linear	model parameter	Disabled: mu*≈0 everywhere; excluded to keep d≤16 (solver limit)
`photo.feedback_on_photosynthesis`	0.5	0.9	linear	model parameter	Disabled: inactive in Farquhar path (light_PAR>20 simple model never called)
`gas_exchange.oxygen_concentration`	210	210	fixed	physical constant	Fixed atmospheric O2; excluded as non-varying
`carbohydrates.starch`	0.05	1	linear	initial condition	Observed range in spinach seedlings (% DW; Geiger & Servaites 1994)